### 4.4.4 Inference on Two Population Proportions

We now consider the case where there are two binomial parameters of interest—say, $p_1$ and $p_2$—and we wish to draw inferences about these proportions. We will present large-sample hypothesis testing and confidence interval procedures based on the normal approximation to the binomial.

***Large-Sample Test for $H_0$: $p_1 = p_2$.*** Suppose that the two independent random samples of sizes $n_1$ and $n_2$ are taken from two populations, and let $x_1$ and $x_2$ represent the number of observations that belong to the class of interest in samples 1 and 2, respectively. Furthermore, suppose that the normal approximation to the binomial is applied to each population, so that the estimators of the population proportions $\hat{p}_1 = x_1/n_1$ and $\hat{p}_2 = x_2/n_2$ and have approximate normal distributions. We are interested in testing the hypotheses

$$H_0: \quad p_1 = p_2$$
$$H_1: \quad p_1 \neq p_2$$

The statistic

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \tag{4.62}$$

is distributed approximately as standard normal and is the basis of a test for $H_0$: $p_1 = p_2$. Specifically, if the null hypothesis $H_0$: $p_1 = p_2$ is true, then using the fact that $p_1 = p_2 = p$, the random variable

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

is distributed approximately $N(0, 1)$. An estimator of the common parameter $p$ is

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

The test statistic for $H_0$: $p_1 = p_2$ is then

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

This leads to the test procedures described here.

**Testing Hypothesis on Two Population Proportions**

Null hypothesis: $H_0$: $p_1 = p_2$

Test statistic:
$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \tag{4.63}$$

| Alternative Hypotheses | Fixed Significance Level Rejection Criterion | *P*-value |
|---|---|---|
| $H_1: p_1 \neq p_2$ | $Z_0 > Z_{\alpha/2}$ or $Z_0 < -Z_{\alpha/2}$ | $P = 2[1 - \Phi(\lvert Z_0 \rvert)]$ |
| $H_1: p_1 > p_2$ | $Z_0 > Z_\alpha$ | $P = 1 - \Phi(Z_0)$ |
| $H_1: p_1 < p_2$ | $Z_0 < -Z_\alpha$ | $P = \Phi(Z_0)$ |

***Confidence Interval on the Difference in Two Population Proportions.*** If there are two population proportions of interest—say, $p_1$ and $p_2$—it is possible to construct a $100(1-\alpha)\%$ CI on their difference. The CI is as follows.

$$\hat{p}_1 - \hat{p}_2 - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \leq p_1 - p_2 \leq \hat{p}_1 - \hat{p}_2 + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \tag{4.64}$$

This result is based on the normal approximation to the **binomial distribution**.

## 4.5 What If There Are More Than Two Populations? The Analysis of Variance

As this chapter has illustrated, testing and experimentation are a natural part of the engineering analysis process and arise often in quality control and improvement problems. Suppose, for example, that an engineer is investigating the effect of different heat-treating methods on the mean hardness of a steel alloy. The experiment would consist of testing several specimens of alloy using each of the proposed heat-treating methods and then measuring the hardness of each specimen. The data from this experiment could be used to determine which heat-treating method should be used to provide maximum mean hardness.

If there are only two heat-treating methods of interest, this experiment could be designed and analyzed using the two-sample *t*-test presented in this chapter. That is, the experimenter has a single **factor** of interest—heat-treating methods—and there are only two **levels** of the factor.

Many single-factor experiments require that more than two levels of the factor be considered. For example, the engineer may want to investigate five different heat-treating methods. In this chapter we show how the **analysis of variance** (ANOVA) can be used for comparing means when there are more than two levels of a single factor. We will also discuss **randomization** of the experimental runs and the important role this concept plays in the overall experimentation strategy. In Part IV, we will discuss how to design and analyze experiments with several factors.

### 4.5.1 An Example

A manufacturer of paper used for making grocery bags is interested in improving the tensile strength of the product. Product engineering thinks that tensile strength is a function of the hardwood concentration in the pulp and that the range of hardwood concentrations of practical

**TABLE 4.4**
**Tensile Strength of Paper (psi)**

| Hardwood Concentration (%) | Observations 1 | 2 | 3 | 4 | 5 | 6 | Totals | Averages |
|---|---|---|---|---|---|---|---|---|
| 5 | 7 | 8 | 15 | 11 | 9 | 10 | 60 | 10.00 |
| 10 | 12 | 17 | 13 | 18 | 19 | 15 | 94 | 15.67 |
| 15 | 14 | 18 | 19 | 17 | 16 | 18 | 102 | 17.00 |
| 20 | 19 | 25 | 22 | 23 | 18 | 20 | 127 | 21.17 |
| | | | | | | | 383 | 15.96 |

interest is between 5% and 20%. A team of engineers responsible for the study decides to investigate four levels of hardwood concentration: 5%, 10%, 15%, and 20%. They decide to make up six test specimens at each concentration level, using a pilot plant. All 24 specimens are tested on a laboratory tensile tester, in random order. The data from this experiment are shown in Table 4.4.

This is an example of a completely randomized single-factor experiment with four levels of the factor. The levels of the factor are sometimes called **treatments,** and each treatment has six observations or **replicates.** The role of **randomization** in this experiment is extremely important. By randomizing the order of the 24 runs, the effect of any nuisance variable that

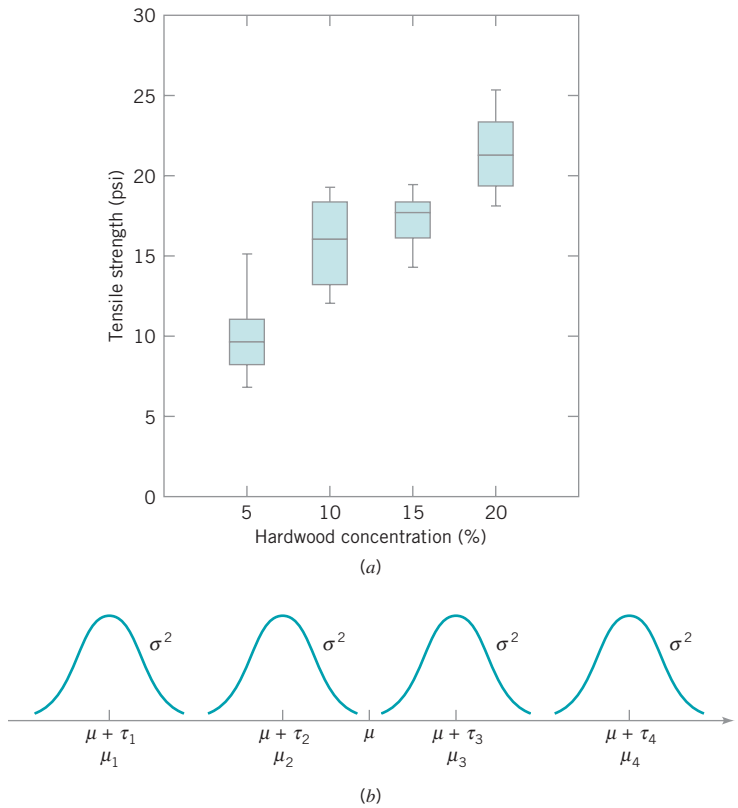



**FIGURE 4.11** (*a*) Box plots of hardwood concentration data. (*b*) Display of the model in equation 4.65 for the completely randomized single-factor experiment.

may influence the observed tensile strength is approximately balanced out. For example, suppose that there is a warm-up effect on the tensile testing machine; that is, the longer the machine is on, the greater the observed tensile strength. If all 24 runs are made in order of increasing hardwood concentration (that is, all six 5% concentration specimens are tested first, followed by all six 10% concentration specimens, etc.), then any observed differences in tensile strength could also be due to the warm-up effect.

It is important to graphically analyze the data from a designed experiment. Figure 4.11*a* presents box plots of tensile strength at the four hardwood concentration levels. This figure indicates that changing the hardwood concentration has an effect on tensile strength; specifically, higher hardwood concentrations produce higher observed tensile strength. Furthermore, the distribution of tensile strength at a particular hardwood level is reasonably symmetric, and the variability in tensile strength does not change dramatically as the hardwood concentration changes.

Graphical interpretation of the data is always useful. Box plots show the variability of the observations *within* a treatment (factor level) and the variability *between* treatments. We now discuss how the data from a single-factor randomized experiment can be analyzed statistically.

### 4.5.2 The Analysis of Variance

Suppose we have $a$ different levels of a single factor that we wish to compare. Sometimes, each factor level is called a treatment, a very general term that can be traced to the early applications of experimental design methodology in the agricultural sciences. The response for each of the $a$ treatments is a random variable. The observed data would appear as shown in Table 4.5. An entry in Table 4.5—say, $y_{ij}$—represents the $j$th observation taken under treatment $i$. We initially consider the case in which there are an equal number of observations, $n$, on each treatment.

We may describe the observations in Table 4.5 by the **linear statistical model**

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1,2,\ldots,a \\ j = 1,2,\ldots,n \end{cases} \tag{4.65}$$

where $y_{ij}$ is a random variable denoting the $(ij)$th observation, $\mu$ is a parameter common to all treatments called the **overall mean,** $\tau_i$ is a parameter associated with the $i$th treatment called the $i$th *treatment effect,* and $\epsilon_{ij}$ is a random error component. Note that the model could have been written as

$$y_{ij} = \mu_i + \varepsilon_{ij} \begin{cases} i = 1,2,\ldots,a \\ j = 1,2,\ldots,n \end{cases}$$

■ **TABLE 4.5**
**Typical Data for a Single-Factor Experiment**

| Treatment | Observations | | | | Totals | Averages |
|---|---|---|---|---|---|---|
| 1 | $y_{11}$ | $y_{12}$ | ... | $y_{1n}$ | $y_{1\cdot}$ | $\bar{y}_{1\cdot}$ |
| 2 | $y_{21}$ | $y_{22}$ | ... | $y_{2n}$ | $y_{2\cdot}$ | $\bar{y}_{2\cdot}$ |
| . | . | . | ... | . | . | . |
| . | . | . | ... | . | . | . |
| . | . | . | ... | . | . | . |
| $a$ | $y_{a1}$ | $y_{a2}$ | ... | $y_{an}$ | $y_{a\cdot}$ | $y_{a\cdot}$ |
| | | | | | $y_{\cdot\cdot}$ | $\bar{y}_{\cdot\cdot}$ |

where $\mu_i = \mu + \tau_i$ is the mean of the $i$th treatment. In this form of the model, we see that each treatment defines a population that has mean $\mu_i$, consisting of the overall mean $\mu$ plus an effect $\tau_i$ that is due to that particular treatment. We will assume that the errors $\varepsilon_{ij}$ are normally and independently distributed with mean zero and variance $\sigma^2$. Therefore, each treatment can be thought of as a normal population with mean $\mu_i$ and variance $\sigma^2$. See Fig. 4.11*b*.

Equation (4.65) is the underlying model for a single-factor experiment. Furthermore, since we require that the observations are taken in random order and that the environment (often called the experimental units) in which the treatments are used is as uniform as possible, this design is called a **completely randomized experimental design.**

We now present the analysis of variance for testing the equality of $a$ population means. This is called a **fixed effects model analysis of variance** (**ANOVA**). However, the ANOVA is a far more useful and general technique; it will be used extensively in Chapters 13 and 14. In this section we show how it can be used to test for equality of treatment effects. The treatment effects $\tau_i$ are usually defined as deviations from the overall mean $\mu$, so that

$$\sum_{i=1}^{a} \tau_i = 0 \tag{4.66}$$

Let $\bar{y}_{i\cdot}$ represent the total of the observations under the $i$th treatment and $\bar{y}_{i\cdot}$ represent the average of the observations under the $i$th treatment. Similarly, let $y_{\cdot\cdot}$ represent the grand total of all observations and $\bar{y}_{\cdot\cdot}$ represent the grand mean of all observations. Expressed mathematically,

$$\begin{aligned} y_{i\cdot} &= \sum_{j=1}^{n} y_{ij} \qquad \bar{y}_{i\cdot} = y_{i\cdot}/n \quad i = 1,2,\ldots,a \\ y_{\cdot\cdot} &= \sum_{i=1}^{a}\sum_{j=1}^{n} y_{ij} \quad \bar{y}_{\cdot\cdot} = y_{\cdot\cdot}/N \end{aligned} \tag{4.67}$$

where $N = an$ is the total number of observations. Thus, the "dot" subscript notation implies summation over the subscript that it replaces.

We are interested in testing the equality of the $a$ treatment means $\mu_1, \mu_2, \ldots, \mu_a$. Using equation (4.66), we find that this is equivalent to testing the hypotheses

$$\begin{aligned} H_0&: \quad \tau_1 = \tau_2 = \cdots = \tau_a = 0 \\ H_1&: \quad \tau_i \neq 0 \text{ for at least one } i \end{aligned} \tag{4.68}$$

Thus, if the null hypothesis is true, each observation consists of the overall mean $\mu$ plus a realization of the random error component $\varepsilon_{ij}$. This is equivalent to saying that all $N$ observations are taken from a normal distribution with mean $\mu$ and variance $\sigma^2$. Therefore, if the null hypothesis is true, changing the levels of the factor has no effect on the mean response.

The ANOVA partitions the total variability in the sample data into two component parts. Then, the test of the hypothesis in equation 4.68 is based on a comparison of two independent estimates of the population variance. The total variability in the data is described by the **total sum of squares**

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n} \left(y_{ij} - \bar{y}_{\cdot\cdot}\right)^2$$

The basic ANOVA partition of the total sum of squares is given in the following definition.

The **sum of squares identity** is

$$\sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{..}\right)^2 = n\sum_{i=1}^{a}\left(\bar{y}_{i.}-\bar{y}_{..}\right)^2 + \sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{i.}\right)^2 \tag{4.69}$$

The proof of this identity is straightforward. Note that we may write

$$\sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{..}\right)^2 = \sum_{i=1}^{a}\sum_{j=1}^{n}\left[\left(\bar{y}_{i.}-\bar{y}_{..}\right)+\left(y_{ij}-\bar{y}_{i.}\right)\right]^2$$

or

$$\sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{..}\right)^2 = n\sum_{i=1}^{a}\left(\bar{y}_{i.}-\bar{y}_{..}\right)^2 + \sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{i.}\right)^2 + 2\sum_{i=1}^{a}\sum_{j=1}^{n}\left(\bar{y}_{i.}-\bar{y}_{..}\right)\left(y_{ij}-\bar{y}_{i.}\right) \tag{4.70}$$

Note that the cross-product term in equation 4.70 is zero, since

$$\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{i.}\right) = y_{i.} - n\bar{y}_{i.} = y_{i.} - n\left(y_{i.}/n\right) = 0$$

Therefore, we have shown that equation 4.70 will reduce to equation (4.69).

The identity in equation (4.69) shows that the total variability in the data, measured by the total sum of squares, can be partitioned into a sum of squares of differences between treatment means and the grand mean and a sum of squares of differences of observations within a treatment from the treatment mean. Differences between observed treatment means and the grand mean measure the differences between treatments, whereas differences of observations within a treatment from the treatment mean can be due only to random error. Therefore, we write equation (4.69) symbolically as

$$SS_T = SS_{\text{Treatments}} + SS_E \tag{4.71}$$

where

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{..}\right)^2 = \text{total sum of squares}$$

$$SS_{\text{Treatments}} = n\sum_{i=1}^{a}\left(\bar{y}_{i.}-\bar{y}_{..}\right)^2 = \text{treatment sum of squares}$$

and

$$SS_E = \sum_{i=1}^{a}\sum_{j=1}^{n}\left(y_{ij}-\bar{y}_{j.}\right)^2 = \text{error sum of squares}$$

We can gain considerable insight into how the ANOVA works by examining the expected values of $SS_{\text{Treatments}}$ and $SS_E$. This will lead us to an appropriate statistic for testing the hypothesis of no differences among treatment means (or $\tau_i = 0$).

The expected value of the treatment sum of squares is

$$E\left(SS_{\text{Treatments}}\right) = (a-1)\sigma^2 + n\sum_{i=1}^{a}\tau_i^2$$

Now if the null hypothesis in equation (4.68) is true, each $\tau_i$ is equal to zero and

$$E\left(\frac{SS_{\text{Treatments}}}{a-1}\right) = \sigma^2$$

If the alternative hypothesis is true, then

$$E\left(\frac{SS_{\text{Treatments}}}{a-1}\right) = \sigma^2 + \frac{n\sum_{i=1}^{a}\tau_i^2}{a-1}$$

The ratio $MS_{\text{Treatments}} = SS_{\text{Treatments}}/(a-1)$ is called the **mean square for treatments.** Thus, if $H_0$ is true, $MS_{\text{Treatments}}$ is an unbiased estimator of $\sigma^2$ whereas if $H_1$ is true, $MS_{\text{Treatments}}$ estimates $\sigma^2$ plus a positive term that incorporates variation due to the systematic difference in treatment means. (Refer to the supplemental material for this chapter for the proofs of these two statements.)

We can also show that the expected value of the error sum of squares is $E(SS_E) = a(n-1)\sigma^2$. Therefore, the **error mean square** $MS_E = SS_E/[a(n-1)]$ is an unbiased estimator of $\sigma^2$ regardless of whether or not $H_0$ is true.

The **error mean square**

$$MS_E = \frac{SS_E}{a(n-1)}$$

is an unbiased estimator of $\sigma^2$.

There is also a partition of the number of degrees of freedom that corresponds to the sum of squares identity in equation (4.69). That is, there are $an = N$ observations; thus, $SS_T$ has $an-1$ degrees of freedom. There are $a$ levels of the factor, so $SS_{\text{Treatments}}$ has $a-1$ degrees of freedom. Finally, within any treatment there are $n$ replicates providing $n-1$ degrees of freedom with which to estimate the experimental error. Since there are $a$ treatments, we have $a(n-1)$ degrees of freedom for error. Therefore, the degrees of freedom partition is

$$an - 1 = a - 1 + a(n-1)$$

Now assume that each of the $a$ populations can be modeled as a normal distribution. Using this assumption we can show that if the null hypothesis $H_0$ is true, the ratio

$$F_0 = \frac{SS_{\text{Treatments}}/(a-1)}{SS_E/[a(n-1)]} = \frac{MS_{\text{Treatments}}}{MS_E} \tag{4.72}$$

has an $F$ distribution with $a - 1$ and $a(n - 1)$ degrees of freedom. Furthermore, from the expected mean squares, we know that $MS_E$ is an unbiased estimator of $\sigma^2$. Also, under the null hypothesis, $MS_{\text{Treatments}}$ is an unbiased estimator of $\sigma^2$. However, if the null hypothesis is false, then the expected value of $MS_{\text{Treatments}}$ is greater than $\sigma^2$. Therefore, under the alternative hypothesis, the expected value of the numerator of the test statistic (equation 4.72) is greater than the expected value of the denominator. Consequently, we should reject $H_0$ if the statistic is large. This implies an upper-tail, one-tail critical region. Therefore, we would reject $H_0$ if $F_0 > F_{\alpha,a-1,a(n-1)}$ where $F_0$ is computed from equation (4.72). A $P$-value approach can also be used, with the $P$-value equal to the probability above $F_0$ in the $F_{a-1,a(n-1)}$ distribution.

Efficient computational formulas for the sums of squares may be obtained by expanding and simplifying the definitions of $SS_{\text{Treatments}}$ and $SS_T$. This yields the following results.

**Definition**

The sums of squares computing formulas for the analysis of variance with equal sample sizes in each treatment are

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{N} \tag{4.73}$$

and

$$SS_{\text{Treatments}} = \sum_{i=1}^{a} \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{N} \tag{4.74}$$

The error sum of squares is obtained by subtraction as

$$SS_E = SS_T - SS_{\text{Treatments}} \tag{4.75}$$

The computations for this test procedure are usually summarized in tabular form as shown in Table 4.6. This is called an **analysis of variance** (or **ANOVA)** table.

■ **TABLE 4.6**
**The Analysis of Variance for a Single-Factor Experiment**

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Square | $F_0$ |
|---|---|---|---|---|
| Treatments | $SS_{\text{Treatments}}$ | $a - 1$ | $MS_{\text{Treatments}}$ | $\frac{MS_{\text{Treatments}}}{MS_E}$ |
| Error | $SS_E$ | $a(n - 1)$ | $MS_E$ | |
| Total | $SS_T$ | $an - 1$ | | |

## EXAMPLE 4.12 The Paper Tensile Strength Experiment

Consider the paper tensile strength experiment described in Section 4.5.1. Use the analysis of variance to test the hypothesis that different hardwood concentrations do not affect the mean tensile strength of the paper.

## SOLUTION

The hypotheses are

$$H_0: \quad \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$$
$$H_1: \quad \tau_i \neq 0 \text{ for at least one } i$$

We will use $\alpha = 0.01$. The sums of squares for the ANOVA are computed from equations 4.73, 4.74, and 4.75 as follows:

$$SS_T = \sum_{i=1}^{4}\sum_{j=1}^{6} y_{ij}^2 - \frac{y_{..}^2}{N}$$

$$= (7)^2 + (8)^2 + \cdots + (20)^2 - \frac{(383)^2}{24} = 512.96$$

$$SS_{\text{Treatments}} = \sum_{i=1}^{4} \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{N}$$

$$= \frac{(60)^2 + (94)^2 + (102)^2 + (127)^2}{6} - \frac{(383)^2}{24} = 382.79$$

$$SS_E = SS_T - SS_{\text{Treatments}}$$
$$= 512.96 - 382.79 = 130.17$$

We usually do not perform these calculations by hand. The ANOVA from Minitab is presented in Table 4.7. Since $F_{0.01,3,20} = 4.94$, we reject $H_0$ and conclude that hardwood concentration in the pulp significantly affects the strength of the paper. Note that the computer output reports a $P$-value for the test statistic $F = 19.61$ in Table 4.7 of zero. This is a truncated value; the actual $P$-value is $P = 3.59 \times 10^{-6}$. However, since the $P$-value is considerably smaller than $\alpha = 0.01$, we have strong evidence to conclude that $H_0$ is not true. Note that Minitab also provides some summary information about each level of hardwood concentration, including the confidence interval on each mean.

**TABLE 4.7**
**Minitab Analysis of Variance Output for the Paper Tensile Strength Experiment**

```
                        One-Way Analysis of Variance

Analysis of Variance
Source     DF        SS        MS        F        P
Factor      3    382.79    127.60    19.61    0.000
Error      20    130.17      6.51
Total      23    512.96

                                      Individual 95% Cls For Mean
                                      Based on Pooled StDev
Level      N      Mean     StDev      —+———+———+———+
5          6    10.000     2.828      (—*—)
10         6    15.667     2.805           (—*—)
15         6    17.000     1.789             (—*—)
20         6    21.167     2.639                  (—*—)
                                      —+———+———+———+
Pooled StDev = 2.551                  10.0    15.0   20.0   25.0
```

Note that the ANOVA tells us whether there is a difference among means. It does not tell us which means differ. If the analysis of variance indicates that there is a statistically significant difference among means, there is a simple graphical procedure that can be used to isolate the specific differences. Suppose that $\bar{y}_{1.}, \bar{y}_{2.}, \ldots, \bar{y}_{a.}$ are the observed averages for these factor levels. Each treatment average has standard deviation $\sigma/\sqrt{n}$, where $\sigma$ is the standard deviation of an individual observation. If all treatment means are equal, the observed means $\bar{y}_{i.}$ would behave as if they were a set of observations drawn at random from a normal distribution with mean $\mu$ and standard deviation $\sigma/\sqrt{n}$.

Visualize this normal distribution capable of being slid along an axis below which the treatment means $\bar{y}_{1.}, \bar{y}_{2.}, \ldots, \bar{y}_{a.}$ are plotted. If all treatment means are equal, there should be some position for this distribution that makes it obvious that the $\bar{y}_{i.}$ values were drawn from the same distribution. If this is not the case, then the $\bar{y}_{i.}$ values that do not appear to have been drawn from this distribution are associated with treatments that produce different mean responses.

The only flaw in this logic is that $\sigma$ is unknown. However, we can use $\sqrt{MS_E}$ from the analysis of variance to estimate $\sigma$. This implies that a $t$ distribution should be used instead of the normal in making the plot, but since the $t$ looks so much like the normal, sketching a normal curve that is approximately $6\sqrt{MS_E/n}$ units wide will usually work very well.

Figure 4.12 shows this arrangement for the hardwood concentration experiment in Section 4.5.1. The standard deviation of this normal distribution is

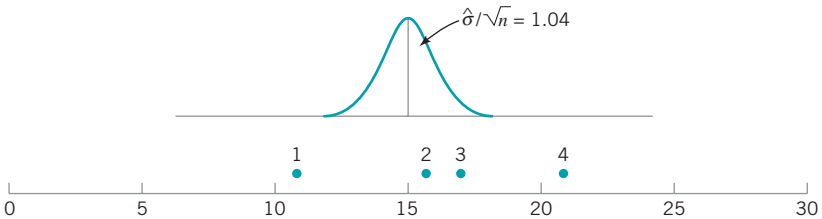
$$\sqrt{MS_E/n} = \sqrt{6.51/6} = 1.04$$

If we visualize sliding this distribution along the horizontal axis, we note that there is no location for the distribution that would suggest that all four observations (the plotted means) are typical, randomly selected values from that distribution. This, of course, should be expected, because the analysis of variance has indicated that the means differ, and the display in Fig. 4.12 is simply a graphical representation of the analysis of variance results. The figure does indicate that treatment 4 (20% hardwood) produces paper with higher mean tensile strength than do the other treatments, and treatment 1 (5% hardwood) results in lower mean tensile strength than do the other treatments. The means of treatments 2 and 3 (10 and 15% hardwood, respectively) do not differ.

This simple procedure is a rough but very useful and effective technique for comparing means following an analysis of variance. However, there are many other more formal ways to do this. For more details on these procedures, see Montgomery (2005).

### 4.5.3 Checking Assumptions: Residual Analysis

The analysis of variance assumes that the model errors (and as a result, the observations) are normally and independently distributed with the same variance in each factor level. These




■ **FIGURE 4.12** Tensile strength averages from the hardwood concentration experiment in relation to a normal distribution with standard deviation $\sqrt{MS_E/n} = \sqrt{6.51/6} = 1.04$

**TABLE 4.8**
**Residuals for the Hardwood Experiment**

| Hardwood Concentration | Residuals | | | | | |
|---|---|---|---|---|---|---|
| 5% | −3.00 | −2.00 | 5.00 | 1.00 | −1.00 | 0.00 |
| 10% | −3.67 | 1.33 | −2.67 | 2.33 | +3.33 | −0.67 |
| 15% | −3.00 | 1.00 | 2.00 | 0.00 | −1.00 | 1.00 |
| 20% | −2.17 | 3.83 | 0.83 | 1.83 | −3.17 | −1.17 |

assumptions can be checked by examining the residuals. We define a residual as the difference between the actual observation $y_{ij}$ and the value $\hat{y}_{ij}$ that would be obtained from a least squares fit of the underlying analysis of variance model to the sample data. For the type of experimental design in this situation, the value $\hat{y}_{ij}$ is the factor-level mean $\bar{y}_{i.}$. Therefore, the residual is $e_{ij} = y_{ij} - \bar{y}_{i.}$; that is, the difference between an observation and the corresponding factor-level mean. The residuals for the hardwood percentage experiment are shown in Table 4.8.

The normality assumption can be checked by constructing a normal probability plot of the residuals. To check the assumption of equal variances at each factor level, plot the residuals against the factor levels and compare the spread in the residuals. It is also useful to plot the residuals against $\bar{y}_{i.}$ (sometimes called the **fitted value**); the variability in the residuals should not depend in any way on the value of $y_{i.}$. When a pattern appears in these plots, it usually suggests the need for data **transformation**—that is, analyzing the data in a different metric. For example, if the variability in the residuals increases with $\bar{y}_{i.}$, then a transformation such as log $y$ or $\sqrt{y}$ should be considered. In some problems the dependency of residual scatter on $y_{i.}$ is very important information. It may be desirable to select the factor level that results in maximum mean response; however, this level may also cause more variation in response from run to run.

The independence assumption can be checked by plotting the residuals against the run order in which the experiment was performed. A pattern in this plot, such as sequences of positive and negative residuals, may indicate that the observations are not independent. This suggests that run order is important or that variables that change over time are important and have not been included in the experimental design.

A normal probability plot of the residuals from the hardwood concentration experiment is shown in Fig. 4.13. Figures 4.14 and 4.15 present the residuals plotted against the

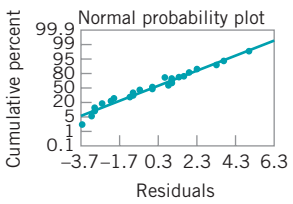



**FIGURE 4.13** Normal probability plot of residuals from the hardwood concentration experiment.

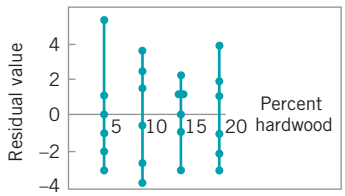



**FIGURE 4.14** Plot of residuals versus factor levels.

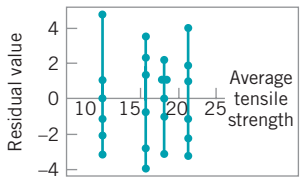



**FIGURE 4.15** Plot of residuals verus $\bar{y}_{i.}$

factor levels and the fitted value $\bar{y}_i$. These plots do not reveal any model inadequacy or unusual problem with the assumptions.

## 4.6 Linear Regression Models

In many problems, two or more variables are related and it is of interest to model and explore this relationship. For example, in a chemical process the yield of product is related to the operating temperature. The chemical engineer may want to build a model relating yield to temperature and then use the model for prediction, process optimization, or process control.

In general, suppose that there is a single **dependent variable** or **response** $y$ that depends on $k$ **independent** or **repressor variables,** for example, $x_1, x_2, \ldots, x_k$ The relationship between these variables is characterized by a mathematical model called a **regression model.** The regression model is fit to a set of sample data. In some instances, the experimenter knows the exact form of the true functional relationship between $y$ and $x_1, x_2, \ldots, x_k$. However, in most cases, the true functional relationship is unknown, and the experimenter chooses an appropriate function to approximate the true model. Low-order polynomial models are widely used as approximating functions.

There are many applications of regression models on quality and process implement. In this section, we present some aspects of fitting these models. More complete presentations of regression are available in Montgomery, Peck, and Vining (2006).

As an example of a linear regression model, suppose that we wish to develop an empirical model relating the viscosity of a polymer to the temperature and the catalyst feed rate. A model that might describe this relationship is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \tag{4.76}$$

where $y$ represents the viscosity, $x_1$ represents the temperature, and $x_2$ represents the catalyst feed rate. This is a **multiple linear regression model** with two independent variables. We often call the independent variables **predictor variables** or **regressors.** The term *linear* is used because Equation 4.76 is a linear function of the unknown parameter $\beta_0$, $\beta_1$, and $\beta_2$. The model describes a plane in the two-dimensional $x_1$, $x_2$, space. The parameter $\beta_0$ defines the intercept of the plane. We sometimes call $\beta_1$ and $\beta_2$ *partial regression coefficients* because $\beta_1$ measures the expected change in $y$ per unit change in $x_1$ when $x_2$ is held constant and $\beta_2$ measures the expected change in $y$ per unit change in $x_2$ when $x_1$ is held constant.

In general, the response variable $y$ may be related to $k$ regressor variables. The model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon \tag{4.77}$$

is called a *multiple linear regression model* with $k$ regressor variables. The parameters $\beta_j, j = 0,1, \ldots, k$, are called the **regression coefficients.** This model describes a hyper plane in the $k$-dimensional space of the regressor variables $\{x_j\}$. The parameter $\beta_j$ represents the expected change in response $y$ per unit change in $x_j$ when all the remaining independent variables $x_i$ $(i \neq j)$ are held constant.

Models that the more complex in appearance than equation (4.77) may often still be analyzed by multiple linear regression techniques. For example, consider adding an interaction term to the first-order model in two variables, say

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon \tag{4.78}$$

# 13 Factorial and Fractional Factorial Experiments for Process Design and Improvement

## Chapter Outline


13.1 WHAT IS EXPERIMENTAL DESIGN?

13.2 EXAMPLES OF DESIGNED EXPERIMENTS IN PROCESS AND PRODUCT IMPROVEMENT

13.3 GUIDELINES FOR DESIGNING EXPERIMENTS

13.4 FACTORIAL EXPERIMENTS
- 13.4.1 An Example
- 13.4.2 Statistical Analysis
- 13.4.3 Residual Analysis

13.5 THE $2^k$ FACTORIAL DESIGN
- 13.5.1 The $2^2$ Design
- 13.5.2 The $2^k$ Design for $k \geq 3$ Factors
- 13.5.3 A Single Replicate of the $2^k$ Design
- 13.5.4 Addition of Center Points to the $2^k$ Design
- 13.5.5 Blocking and Confounding in the $2^k$ Design

13.6 FRACTIONAL REPLICATION OF THE $2^k$ DESIGN
- 13.6.1 The One-Half Fraction of the $2^k$ Design
- 13.6.2 Smaller Fractions: The $2^{k-p}$ Fractional Factorial Design

**Supplemental Material for Chapter 13**

- S13.1 Additional Discussion of Guidelines for Planning Experiments
- S13.2 Using a *t*-Test for Detecting Curvature
- S13.3 Blocking in Designed Experiments
- S13.4 More about Expected Mean Squares in the Analysis of Variance


The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

## Chapter Overview and Learning Objectives

Most experiments for process design and improvement involve several variables. Factorial experimental designs, and their variations, are used in such situations. This chapter gives an introduction to factorial designs, emphasizing their applications for process and quality improvement. Designed experiments are widely used in the DMAIC process, and are an extremely important in the improve step. They also play a leading role in design for six-sigma (DFSS). We concentrate on experimental designs where all the factors have two levels, and show how fractional versions of these designs can be used with great effectiveness in industrial experimentation. Important topics include the analysis of factorial experimental designs and the use of graphical methods in interpretation of the results. Both the interaction graph and a response surface plot are shown to be very useful in interpretation of results.

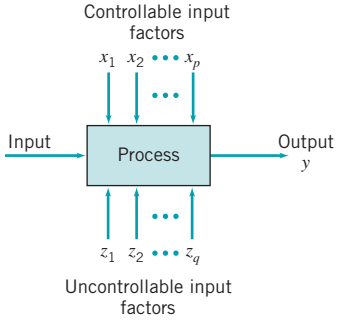
After careful study of this chapter you should be able to do the following:

1. Explain how designed experiments can be used to improve product design and improve process performance
2. Explain how designed experiments can be used to reduce the cycle time required to develop new products and processes
3. Understand how main effects and interactions of factors can be estimated
4. Understand the factorial design concept
5. Know how to use the analysis of variance (ANOVA) to analyze data from factorial designs
6. Know how residuals are used for model adequacy checking for factorial designs
7. Know how to use the $2^k$ system of factorial designs
8. Know how to construct and interpret contour plots and response surface plots
9. Know how to add center points to a $2^k$ factorial design to test for curvature and provide an estimate of pure experimental error
10. Understand how the blocking principal can be used in a factorial design to eliminate the effects of a nuisance factor
11. Know how to use the $2^{k-p}$ system of fractional factorial designs

## 13.1 What Is Experimental Design?

As indicated in Chapter 1, a designed experiment is a test or series of tests in which purposeful changes are made to the input variables of a process so that we may observe and identify corresponding changes in the output response. The process, as shown in Fig. 13.1, can be visualized as some combination of machines, methods, and people that transforms an input material into an output product. This output product has one or more observable quality characteristics or responses. Some of the process variables $x_1, x_2, \ldots, x_p$ are **controllable,** whereas others $z_1, z_2, \ldots, z_q$ are **uncontrollable** (although they may be controllable for purposes of the test). Sometimes these uncontrollable factors are called **noise** factors. The objectives of the experiment may include

1. Determining which variables are most influential on the response, $y$.
2. Determining where to set the influential $x$'s so that $y$ is near the nominal requirement.
3. Determining where to set the influential $x$'s so that variability in $y$ is small.
4. Determining where to set the influential $x$'s so that the effects of the uncontrollable variables $z$ are minimized.




■ **FIGURE 13.1** General model of a process.

Thus, experimental design methods may be used either in process development or process troubleshooting to improve process performance or to obtain a process that is **robust** or **insensitive** to external sources of variability.

Statistical process-control methods and experimental design, two very powerful tools for the improvement and optimization of processes, are closely interrelated. For example, if a process is in statistical control but still has poor capability, then to improve process capability it will be necessary to reduce variability. Designed experiments may offer a more effective way to do this than SPC. Essentially, SPC is a **passive** statistical method: We watch the process and wait for some information that will lead to a useful change. However, if the process is in control, passive observation may not produce much useful information. On the other hand, experimental design is an **active** statistical method: We will actually perform a series of tests on the process or system, making changes in the inputs and observing the corresponding changes in the outputs, and this will produce information that can lead to process improvement.

Experimental design methods can also be very useful in establishing statistical control of a process. For example, suppose that a control chart indicates that the process is out of control, and the process has many controllable input variables. Unless we know *which* input variables are the important ones, it may be very difficult to bring the process under control. Experimental design methods can be used to identify these influential process variables.

Experimental design is a critically important engineering tool for improving a manufacturing process. It also has extensive application in the development of new processes. Application of these techniques early in process development can result in

1. Improved yield
2. Reduced variability and closer conformance to the nominal
3. Reduced development time
4. Reduced overall costs

Experimental design methods can also play a major role in **engineering design** activities, where new **products** are developed and existing ones improved. Designed experiments are widely used in design for six-sigma (DFSS) activities. Some applications of statistical experimental design in engineering design include

1. Evaluation and comparison of basic design configurations
2. Evaluation of material alternatives
3. Determination of key product design parameters that impact performance

Use of experimental design in these areas can result in improved manufacturability of the product, enhanced field performance and reliability, lower product cost, and shorter product development time.

## 13.2 Examples of Designed Experiments in Process and Product Improvement

In this section, we present several examples that illustrate the application of designed experiments in improving process and product quality. In subsequent sections, we will demonstrate the statistical methods used to analyze the data and draw conclusions from experiments such as these.

### EXAMPLE 13.1 Characterizing a Process

An engineer has applied SPC to a process for soldering electronic components to printed circuit boards. Through the use of *u* charts and Pareto analysis he has established statistical control of the flow solder process and has reduced the average number of defective solder joints per board to around 1%. However, since the average board contains over 2000 solder joints, even 1% defective presents far too many solder joints requiring rework. The engineer would like to reduce defect levels even further; however, since the process is in statistical control, it is not obvious what machine adjustments will be necessary.

The flow solder machine has several variables that can be controlled. They include

1. Solder temperature
2. Preheat temperature
3. Conveyor speed
4. Flux type
5. Flux specific gravity
6. Solder wave depth
7. Conveyor angle

In addition to these controllable factors, several others cannot be easily controlled during routine manufacturing, although they could be controlled for purposes of a test. They are

1. Thickness of the printed circuit board
2. Types of components used on the board
3. Layout of the components on the board
4. Operator
5. Production rate

In this situation, the engineer is interested in **characterizing** the flow solder machine; that is, he wants to determine which factors (both controllable and uncontrollable) affect the occurrence of defects on the printed circuit boards. To accomplish this task he can design an experiment that will enable him to estimate the magnitude and direction of the factor effects. That is, how much does the response variable (defects per unit) change when each factor is changed, and does changing the factors *together* produce different results than are obtained from individual factor adjustments? A factorial experiment will be required to do this. Sometimes we call this kind of factorial experiment a **screening experiment.**

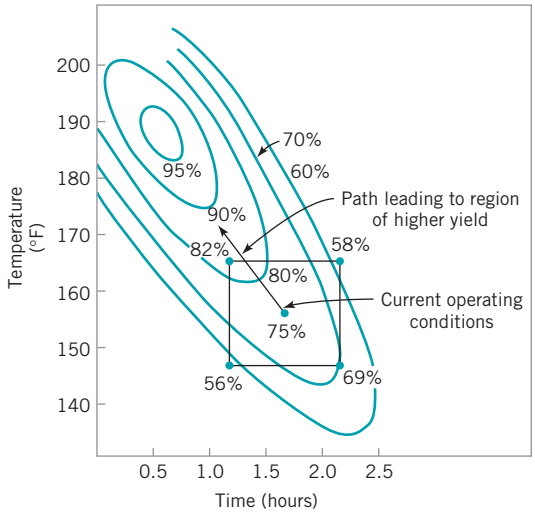
The information from this screening or characterization experiment will be used to identify the critical process factors and to determine the direction of adjustment for these factors to further reduce the number of defects per unit. The experiment may also provide information about which factors should be more carefully controlled during routine manufacturing to prevent high defect levels and erratic process performance. Thus, one result of the experiment could be the application of control charts to one or more *process* variables (such as solder temperature) in addition to the *u* chart on process output. Over time, if the process is sufficiently improved, it may be possible to base most of the process-control plan on controlling process input variables instead of control charting the output.

### EXAMPLE 13.2 Optimizing a Process

In a characterization experiment, we are usually interested in determining which process variables affect the response. A logical next step is to **optimize**—that is, to determine the region in the important factors that lead to the best possible response. For example, if the response is yield, we will look for a region of maximum yield, and if the response is variability in a critical product dimension, we will look for a region of minimum variability.

Suppose we are interested in improving the yield of a chemical process. Let's say that we know from the results of a characterization experiment that the two most important process variables that influence yield are operating temperature and reaction time. The process currently runs at 155° F and 1.7 h of reaction time, producing yields around 75%. Figure 13.2 shows a view of the time-temperature region from above. In this graph the lines of constant yield are connected to form **response contours,** and we have shown the contour lines for 60, 70, 80, 90, and 95% yield.

To locate the optimum, it is necessary to perform an experiment that varies time and temperature together. This type of experiment is called a **factorial experiment;** an example of a




factorial experiment with both time and temperature run at two levels is shown in Fig. 13.2. The responses observed at the four corners of the square indicate that we should move in the general direction of increased temperature and decreased reaction time to increase yield. A few additional runs could be performed in this direction, which would be sufficient to locate the region of maximum yield. Once we are in the region of the optimum, a more elaborate experiment could be performed to give a very precise estimate of the optimum operating condition. This type of experiment, called a **response surface experiment,** is discussed in Chapter 14.

■ **FIGURE 13.2** Contour plot of yield as a function of reaction time and reaction temperature, illustrating an optimization experiment.

## EXAMPLE 13.3 A Product Design Example

Designed experiments can often be applied in the product design process. To illustrate, suppose that a group of engineers are designing a door hinge for an automobile. The quality characteristic of interest is the check effort, or the holding ability of the door latch that prevents the door from swinging closed when the vehicle is parked on a hill. The check mechanism consists of a spring and a roller. When the door is opened, the roller travels through an arc causing the leaf spring to be compressed. To close the door, the spring must be forced aside, which creates the check effort. The engineering team believes the check effort is a function of the following factors:

1. Roller travel distance
2. Spring height pivot to base
3. Horizontal distance from pivot to spring
4. Free height of the reinforcement spring
5. Free height of the main spring

The engineers build a prototype hinge mechanism in which all these factors can be varied over certain ranges. Once appropriate levels for these five factors are identified, an experiment can be designed consisting of various combinations of the factor levels, and the prototype hinge can be tested at these combinations. This will produce information concerning which factors are most influential on latch check effort, and through use of this information the design can be improved.

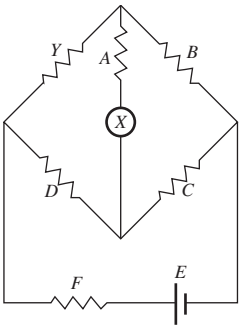
## EXAMPLE 13.4 Determining System and Component Tolerances

The Wheatstone bridge shown in Fig. 13.3 is a device used for measuring an unknown resistance, $Y$. The adjustable resistor $B$ is manipulated until a particular current flow is obtained through the ammeter (usually $X = 0$). Then the unknown resistance is calculated as

$$Y = \frac{BD}{C} - \frac{X^2}{C^2 E}\left[A(D+C) + D(B+C)\right]\left[B(C+D) + F(B+C)\right] \tag{13.1}$$

The engineer wants to design the circuit so that overall gauge capability is good; that is, he would like the standard deviation of measurement error to be small. He has decided that $A = 20\ \Omega$, $C = 2\ \Omega$, $D = 50\ \Omega$, $E = 1.5\ \Omega$, and $F = 2\ \Omega$ is the best choice of the design parameters as far as gauge capability is concerned, but the overall measurement error is still too high. This is likely due to the tolerances that have been specified on the circuit components. These tolerances are ±1% for each resistor $A$, $B$, $C$, $D$, and $F$, and ±5% for the

(*continued*)




■ FIGURE 13.3 A Wheatstone bridge.

power supply $E$. These tolerance bands can be used to define appropriate factor levels, and an experiment can be performed to determine which circuit components have the most critical tolerances and how much they must be tightened to produce adequate gauge capability. The information from this experiment will result in a design specification that tightens only the most critical tolerances the minimum amount possible consistent with desired measurement capability. Consequently, a lower cost design that is easier to manufacture will be possible.

Notice that in this experiment it is unnecessary actually to build hardware, since the response from the circuit can be calculated via equation (13.1). The actual response variable for the experiment should be the standard deviation of $Y$. However, an equation for the transmitted variation in $Y$ from the circuit can be found using the methods of Section 8.6.2. Therefore, the entire experiment can be performed using a computer model of the Wheatstone bridge.

## 13.3 Guidelines for Designing Experiments

Designed experiments are a powerful approach to improving a process. To use this approach, it is necessary that everyone involved in the experiment have a clear idea in advance of the objective of the experiment, exactly what factors are to be studied, how the experiment is to be conducted, and at least a qualitative understanding of how the data will be analyzed. Montgomery (2005) gives an outline of the recommended procedure, reproduced in Fig. 13.4. We now briefly amplify each point in this checklist.

**1. Recognition of and statement of the problem.** In practice, it is often difficult to realize that a problem requiring formal designed experiments exists, so it may not be easy to develop a clear and generally accepted statement of the problem. However, it is absolutely essential to fully develop all ideas about the problem and about the specific objectives of the experiment. Usually, it is important to solicit input from all concerned parties—engineering, quality, marketing, the customer, management, and the operators (who usually have much insight that is all too often ignored). A clear statement of the problem and the objectives of the experiment often contributes substantially to better process understanding and eventual solution of the problem.

Pre-experimental planning
1. Recognition of and statement of the problem
2. Choice of factors and levels
3. Selection of the response variable

(2, 3: often done simultaneously, or in reverse order)

4. Choice of experimental design
5. Performing the experiment
6. Data analysis
7. Conclusions and recommendations

■ FIGURE 13.4 Procedure for designing an experiment.

**2. Choice of factors and levels.** The experimenter must choose the factors to be varied in the experiment, the ranges over which these factors will be varied, and the specific levels at which runs will be made. Process knowledge is required to do this. This process knowledge is usually a combination of practical experience and theoretical understanding. It is important to investigate all factors that may be of importance and to avoid being overly influenced by past experience, particularly when we are in the early stages of experimentation or when the process is not very mature. When the objective is factor screening or process characterization, it is usually best to keep the number of factor levels low. (Most often two levels are used.) As noted in Fig. 13.4, steps 2 and 3 are often carried out simultaneously, or step 3 may be done first in some applications.

**3. Selection of the response variable.** In selecting the response variable, the experimenter should be certain that the variable really provides useful information about the process under study. Most often the average or standard deviation (or both) of the measured characteristic will be the response variable. Multiple responses are not unusual. Gauge capability is also an important factor. If gauge capability is poor, then only relatively large factor effects will be detected by the experiment, or additional replication will be required.

**4. Choice of experimental design.** If the first three steps are done correctly, this step is relatively easy. Choice of design involves consideration of sample size (number of replicates), selection of a suitable run order for the experimental trials, and whether or not blocking or other randomization restrictions are involved. This chapter and Chapter 14 illustrate some of the more important types of experimental designs.

**5. Performing the experiment.** When running the experiment, it is vital to carefully monitor the process to ensure that everything is being done according to plan. Errors in experimental procedure at this stage will usually destroy experimental validity. Up-front planning is crucial to success. It is easy to underestimate the logistical and planning aspects of running a designed experiment in a complex manufacturing environment.

**6. Data analysis.** Statistical methods should be used to analyze the data so that results and conclusions are objective rather than judgmental. If the experiment has been designed correctly and if it has been performed according to the design, then the type of statistical method required is not elaborate. Many excellent software packages are available to assist in the data analysis, and simple graphical methods play an important role in data interpretation. Residual analysis and model validity checking are also important.

**7. Conclusions and recommendations.** Once the data have been analyzed, the experiment must draw *practical* conclusions about the results and recommend a course of action. Graphical methods are often useful in this stage, particularly in presenting the results to others. Follow-up runs and confirmation testing should also be performed to validate the conclusions from the experiment.

Steps 1 to 3 are usually called **pre-experimental planning.** It is vital that these steps be performed as well as possible if the experiment is to be successful. Coleman and Montgomery (1993) discuss this in detail and offer more guidance in pre-experimental planning, including worksheets to assist the experimenter in obtaining and documenting the required information. Section S13.1 of the supplemental text material contains additional useful material on planning experiments.

Throughout this entire process, it is important to keep in mind that experimentation is an important part of the learning process, where we tentatively formulate hypotheses about a system, perform experiments to investigate these hypotheses, and on the basis of the results formulate new hypotheses, and so on. This suggests that experimentation is **iterative**. It is usually a major mistake to design a single, large comprehensive experiment at the start of a

study. A successful experiment requires knowledge of the important factors, the ranges over which these factors should be varied, the appropriate number of levels to use, and the proper units of measurement for these variables. Generally, we do not know perfectly the answers to these questions, but we learn about them as we go along. As an experimental program progresses, we often drop some variables, add others, change the region of exploration for some factors, or add new response variables. Consequently, we usually experiment **sequentially,** and as a general rule, no more than about 25% of the available resources should be invested in the first experiment. This will ensure that sufficient resources are available to accomplish the final objective of the experiment.

## 13.4 Factorial Experiments

When there are several factors of interest in an experiment, a **factorial design** should be used. In such designs factors are varied together. Specifically, by a factorial experiment we mean that in each complete trial or replicate of the experiment all possible combinations of the levels of the factors are investigated. Thus, if there are two factors $A$ and $B$ with $a$ levels of factor $A$ and $b$ levels of factor $B$, then each replicate contains all $ab$ possible combinations.

The effect of a factor is defined as the change in response produced by a change in the level of the factor. This is called a **main effect** because it refers to the primary factors in the study. For example, consider the data in Fig. 13.5. In this factorial design, both the factors $A$ and $B$ have two levels, denoted by "–" and "+." These two levels are called "low" and "high," respectively. The main effect of factor $A$ is the difference between the average response at the high level of $A$ and the average response at the low level of $A$, or

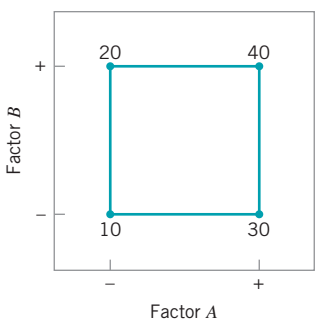
$$A = \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{30+40}{2} - \frac{10+20}{2} = 20$$

That is, changing factor $A$ from the low level (–) to the high level (+) causes an average response increase of 20 units. Similarly, the main effect of $B$ is

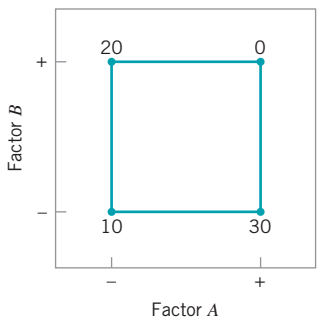
$$B = \bar{y}_{B^+} - \bar{y}_{B^-} = \frac{20+40}{2} - \frac{10+30}{2} = 10$$

In some experiments, the difference in response between the levels of one factor is not the same at all levels of the other factors. When this occurs, there is an **interaction** between the factors. For example, consider the data in Fig. 13.6. At the low level of factor $B$, the $A$ effect is

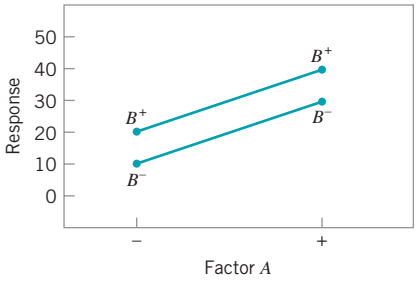
$$A = 30 - 10 = 20$$



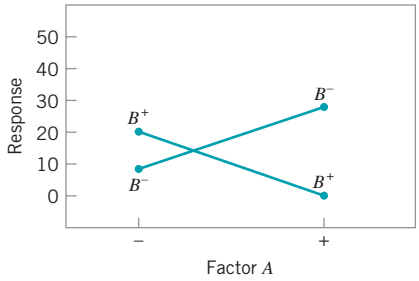

■ FIGURE 13.5 A factorial experiment with two factors.




■ FIGURE 13.6 A factorial experiment with interaction.




■ **FIGURE 13.7** Factorial experiment, no interaction.




■ **FIGURE 13.8** Factorial experiment with interaction.

and at the high level of factor $B$, the $A$ effect is

$$A = 0 - 20 = -20$$

Since the effect of $A$ depends on the level chosen for factor $B$, there is interaction between $A$ and $B$.

When an interaction is large, the corresponding main effects have little meaning. For example, by using the data in Fig. 13.6, we find the main effect of $A$ as

$$A = \frac{30+0}{2} - \frac{10+20}{2} = 0$$

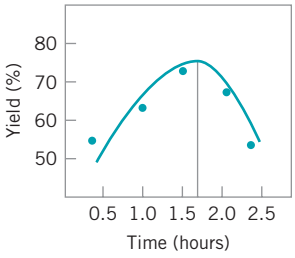
and we would be tempted to conclude that there is no $A$ effect. However, when we examine the main effect of $A$ at *different levels of factor B*, we see that this is not the case. The effect of factor $A$ depends on the levels of factor $B$. Thus, knowledge of the $AB$ interaction is more useful than knowledge of the main effect. A significant interaction can mask the significance of main effects.

The concept of interaction can be illustrated graphically. Figure 13.7 plots the data in Fig. 13.5 against the levels of $A$ for both levels of $B$. Note that the $B^-$ and $B^+$ lines are roughly parallel, indicating that factors $A$ and $B$ do not interact. Figure 13.8 plots the data in Fig. 13.6. In Fig. 13.8, the $B^-$ and $B^+$ lines are not parallel, indicating the interaction between factors $A$ and $B$. Such graphical displays are often useful in presenting the results of experiments.

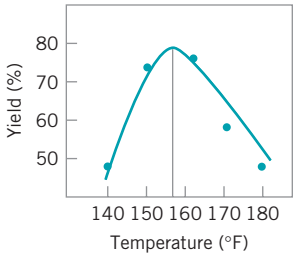
An alternative to the factorial design that is (unfortunately) used in practice is to change the factors one at a time rather than to vary them simultaneously. To illustrate the one-factor-at-a-time procedure, consider the optimization experiment described earlier in Example 13.2. The engineer is interested in finding the values of temperature and time that maximize yield. Suppose that we fix temperature at 155°F (the current operating level) and perform five runs at different levels of time—say, 0.5 h, 1.0 h, 1.5 h, 2.0 h, and 2.5 h. The results of this series of runs are shown in Fig. 13.9. This figure indicates that maximum yield is achieved at about 1.7 h of reaction time. To optimize temperature, the engineer fixes time at 1.7 h (the apparent optimum) and performs five runs at different temperatures—say, 140°F, 150°F, 160°F, 170°F, and 180°F. The results of this set of runs are plotted in Fig. 13.10. Maximum yield occurs at about 155°F. Therefore, we would conclude that running the process at 155°F and 1.7 h is the best set of operating conditions, resulting in yields around 75%.

Figure 13.11 displays the contour plot of yield as a function of temperature and time with the one-factor-at-a-time experiment shown on the contours. Clearly, the one-factor-at-a-time design has failed dramatically here, as the true optimum is at least 20 yield points higher and occurs at much lower reaction times and higher temperatures. The failure to discover the shorter reaction times is particularly important because it could have significant impact on production volume or capacity, production planning, manufacturing cost, and total productivity.

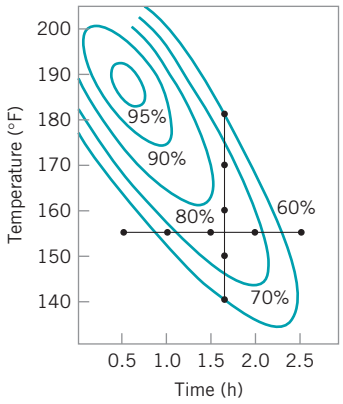
The one-factor-at-a-time method has failed here because it fails to detect the interaction between temperature and time. Factorial experiments are the only way to detect interactions.




■ **FIGURE 13.9** Yield versus reaction time with temperature constant at 155°F.




■ **FIGURE 13.10** Yield versus temperature with reaction time constant at 1.7 h.




■ **FIGURE 13.11** Optimization experiment using the one-factor-at-a-time method.

Furthermore, the one-factor-at-a-time method is inefficient; it will require more experimentation than a factorial, and as we have just seen, there is no assurance that it will produce the correct results. The experiment shown in Fig. 13.2 that produced the information pointing to the region of the optimum is a simple example of a factorial experiment.

### 13.4.1 An Example

Aircraft primer paints are applied to aluminum surfaces by two methods—dipping and spraying. The purpose of the primer is to improve paint adhesion; some parts can be primed using either application method. A team using the DMAIC approach has identified three different primers that can be used with both application methods. Three specimens were painted with each primer using each application method, a finish paint was applied, and the adhesion force was measured. The 18 runs from this experiment were run in random order. The resulting data are shown in Table 13.1. The circled numbers in the cells are the cell totals. The objective of the experiment was to determine which combination of primer paint and application method produced the highest adhesion force. It would be desirable if at least one of the primers produced high adhesion force *regardless* of application method, as this would add some flexibility to the manufacturing process.

### 13.4.2 Statistical Analysis

The analysis of variance (ANOVA) described in Chapter 4 can be extended to handle the two-factor factorial experiment. Let the two factors be denoted $A$ and $B$, with $a$ levels of factor $A$ and

■ **TABLE 13.1**
**Adhesion Force Data**

| | Application Method | | | | |
|---|---|---|---|---|---|
| **Primer Type** | **Dipping** | | **Spraying** | | $y_{i..}$ |
| 1 | 4.0, 4.5, 4.3 | (12.8) | 5.4, 4.9, 5.6 | (15.9) | 28.7 |
| 2 | 5.6, 4.9, 5.4 | (15.9) | 5.8, 6.1, 6.3 | (18.2) | 34.1 |
| 3 | 3.8, 3.7, 4.0 | (11.5) | 5.5, 5.0, 5.0 | (15.5) | 27.0 |
| $y_{.j.}$ | 40.2 | | 49.6 | | $89.8 = y_{...}$ |

**TABLE 13.2**
**Data for a Two-Factor Factorial Design**

| | Factor *B* | | | |
|---|---|---|---|---|
| | **1** | **2** | **…** | ***b*** |
| 1 | $y_{111}, y_{112}, \ldots, y_{11n}$ | $y_{121}, y_{122}, \ldots, y_{12n}$ | … | $y_{1b1}, y_{1b2}, \ldots, y_{1bn}$ |
| 2 | $y_{211}, y_{212}, \ldots, y_{21n}$ | $y_{221}, y_{222}, \ldots, y_{22n}$ | … | $y_{2b1}, y_{2b2}, \ldots, y_{2bn}$ |
| Factor *A* | | | | |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| *a* | $y_{a11}, y_{a12}, \ldots, y_{a1n}$ | $y_{a21}, y_{a22}, \ldots, y_{a2n}$ | … | $y_{ab1}, y_{ab2}, \ldots, y_{abn}$ |

*b* levels of *B*. If the experiment is replicated *n* times, the data layout will look like Table 13.2. In general, the observation in the *ij*th cell in the *k*th replicate is $y_{ijk}$. In collecting the data, the *abn* observations would be run in *random* order. Thus, like the single-factor experiment studied in Chapter 4, the two-factor factorial is a **completely randomized design.** Both factors are assumed to be fixed effects.

The observations from a two-factor factorial experiment may be described by the model

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \quad \begin{cases} i = 1,2,\ldots,a \\ j = 1,2,\ldots,b \\ k = 1,2,\ldots,n \end{cases} \tag{13.2}$$

where $\mu$ is the overall mean effect, $\tau_i$ is the effect of the *i*th level of factor *A*, $\beta_j$ is the effect of the *j*th level of factor *B*, $(\tau\beta)_{ij}$ is the effect of the interaction between *A* and *B*, and $\varepsilon_{ijk}$ is an NID(0, $\sigma^2$) random error component. We are interested in testing the hypotheses of no significant factor *A* effect, no significant factor *B* effect, and no significant *AB* interaction.

Let $y_{i..}$ denote the total of the observations at the *i*th level of factor *A*, $y_{.j.}$ denote the total of the observations at the *j*th level of factor *B*, $y_{ij.}$ denote the total of the observations in the *ij*th cell of Table 13.2, and $y_{...}$ denote the grand total of all the observations.

Define $\bar{y}_{i..}$, $\bar{y}_{.j.}$, $\bar{y}_{ij.}$, and $\bar{y}_{...}$ as the corresponding row, column, cell, and grand averages. That is,

$$\begin{aligned}
y_{i..} &= \sum_{j=1}^{b}\sum_{k=1}^{n} y_{ijk} & \bar{y}_{i..} &= \frac{y_{i..}}{bn} & i &= 1,2,\ldots,a \\
y_{.j.} &= \sum_{i=1}^{a}\sum_{k=1}^{n} y_{ijk} & \bar{y}_{.j.} &= \frac{y_{.j.}}{an} & j &= 1,2,\ldots,b \\
y_{ij.} &= \sum_{k=1}^{n} y_{ijk} & \bar{y}_{ij.} &= \frac{y_{ij.}}{n} & i &= 1,2,\ldots,a \\
& & & & j &= 1,2,\ldots,b \\
y_{...} &= \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n} y_{ijk} & \bar{y}_{...} &= \frac{y_{...}}{abn}
\end{aligned} \tag{13.3}$$

The analysis of variance decomposes the total corrected sum of squares

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}\left(y_{ijk} - \bar{y}_{...}\right)^2$$

as follows:

$$\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}\left(y_{ijk} - \bar{y}_{...}\right)^2 = bn\sum_{i=1}^{a}\left(\bar{y}_{i..} - \bar{y}_{...}\right)^2 + an\sum_{j=1}^{b}\left(\bar{y}_{.j.} - \bar{y}_{...}\right)^2 + n\sum_{i=1}^{a}\sum_{j=1}^{b}\left(\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}\right)^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n}\left(y_{ijk} - \bar{y}_{ij.}\right)^2$$

or symbolically,

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E \tag{13.4}$$

The corresponding degree of freedom decomposition is

$$abn - 1 = (a-1) + (b-1) + (a-1)(b-1) + ab(n-1) \tag{13.5}$$

This decomposition is usually summarized in an analysis of variance table such as the one shown in Table 13.3.

To test for no row factor effects, no column factor effects, and no interaction effects, we would divide the corresponding mean square by mean square error. Each of these ratios will follow an $F$ distribution, with numerator degrees of freedom equal to the number of degrees of freedom for the numerator mean square and $ab(n - 1)$ denominator degrees of freedom, when the null hypothesis of no factor effect is true. We would reject the corresponding

■ **TABLE 13.3**
**The ANOVA Table for a Two-Factor Factorial, Fixed Effects Model**

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Square | $F_0$ |
|---|---|---|---|---|
| $A$ | $SS_A$ | $a-1$ | $MS_A = \frac{SS_A}{a-1}$ | $F_0 = \frac{MS_A}{MS_E}$ |
| $B$ | $SS_B$ | $b-1$ | $MS_B = \frac{SS_B}{b-1}$ | $F_0 = \frac{MS_B}{MS_E}$ |
| Interaction | $SS_{AB}$ | $a-1\ \ b-1$ | $MS_{AB} = \frac{SS_{AB}}{a-1\ \ b-1}$ | $F_0 = \frac{MS_{AB}}{MS_E}$ |
| Error | $SS_E$ | $ab\ \ n-1$ | $MS_E = \frac{SS_E}{ab\ \ n-1}$ | |
| Total | $SS_T$ | $abn-1$ | | |

hypothesis if the computed $F$ exceeded the tabular value at an appropriate significance level, or alternatively, if the $P$-value were smaller than the specified significance level.

The ANOVA is usually performed with computer software, although simple computing formulas for the sums of squares may be obtained easily. The computing formulas for these sums of squares follow.

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n} y_{ijk}^2 - \frac{y_{...}^2}{abn} \tag{13.6}$$

**Main effects**

$$SS_A = \sum_{i=1}^{a} \frac{y_{i..}^2}{bn} - \frac{y_{...}^2}{abn} \tag{13.7}$$

$$SS_B = \sum_{j=1}^{b} \frac{y_{.j.}^2}{an} - \frac{y_{...}^2}{abn} \tag{13.8}$$

**Interaction**

$$SS_{AB} = \sum_{i=1}^{a}\sum_{j=1}^{b} \frac{y_{ij.}^2}{n} - \frac{y_{...}^2}{abn} - SS_A - SS_B \tag{13.9}$$

**Error**

$$SS_E = SS_T - SS_A - SS_B - SS_{AB} \tag{13.10}$$

## EXAMPLE 13.5 The Aircraft Primer Paint Problem

Use the ANOVA described above to analyze the aircraft primer paint experiment described in Section 13.4.1.

### SOLUTION

The sums of squares required are

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n} y_{ijk}^2 - \frac{y_{...}^2}{abn}$$

$$= (4.0)^2 + (4.5)^2 + \cdots + (5.0)^2 - \frac{(89.8)^2}{18} = 10.72$$

$$SS_{\text{primers}} = \sum_{i=1}^{a} \frac{y_{i..}^2}{bn} - \frac{y_{...}^2}{abn}$$

$$= \frac{(28.7)^2 + (34.1)^2 + (27.0)^2}{6} - \frac{(89.8)^2}{18} = 4.58$$

$$SS_{\text{methods}} = \sum_{j=1}^{b} \frac{y_{.j.}^2}{an} - \frac{y_{...}^2}{abn}$$

$$= \frac{(40.2)^2 + (49.6)^2}{9} - \frac{(89.8)^2}{18} = 4.91$$

$$SS_{\text{interaction}} = \sum_{i=1}^{a}\sum_{j=1}^{b} \frac{y_{ij.}^2}{n} - \frac{y_{...}^2}{abn} - SS_{\text{primers}} - SS_{\text{methods}}$$

$$= \frac{(12.8)^2 + (15.9)^2 + (11.5)^2 + (15.9)^2 + (18.2)^2 + (15.5)^2}{3}$$

$$- \frac{(89.8)^2}{18} - 4.58 - 4.91 = 0.24$$

and

$$SS_E = SS_T - SS_{\text{primers}} - SS_{\text{methods}} - SS_{\text{interaction}}$$

$$= 10.72 - 4.58 - 4.91 - 0.24 = 0.99$$

The ANOVA is summarized in Table 13.4. Note that the $P$-values for both main effects are very small, indicating that the type of primer used and the application method significantly affect adhesion force. Since the $P$-value for the interaction effect $F$-ratio is relatively large, we would conclude that there

(*continued*)

■ TABLE 13.4
**ANOVA for Example 13.5**

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Square | $F_0$ | $P$-value |
|---|---|---|---|---|---|
| Primer types | 4.58 | 2 | 2.290 | 27.93 | $1.93 \times 10^{-4}$ |
| Application methods | 4.91 | 1 | 4.910 | 59.88 | $5.28 \times 10^{-6}$ |
| Interaction | 0.24 | 2 | 0.120 | 1.46 | 0.269 |
| Error | 0.99 | 12 | 0.082 | | |
| Total | 10.72 | 17 | | | |

is no interaction between primer type and application method. As an alternative to using the $P$-values, we could compare the computed $F$-ratios to a 5% (say) upper critical value of the $F$ distribution. Since $F_{0.05,2,12} = 3.89$ and $F_{0.05,1,12} = 4.75$, we conclude that primer type and application method affect adhesion force. Furthermore, since $1.5 < F_{0.05,2,12}$, there is no indication of interaction between these factors.

In practice, ANOVA computations are performed on a computer using a statistics software package. Table 13.5 is the analysis of variance from Minitab. Note the similarity of this display to Table 13.4. Because the computer carries more decimal places than we did in the manual calculations, the $F$-ratios in Tables 13.4 and 13.5 are slightly different. The $P$-value for each $F$-ratio is called "significance level" in Table 13.5, and when a $P$-value is less than 0.001, Minitab reports it as 0.000.

A graph of the adhesion force cell averages $\{\bar{y}_{ij.}\}$ versus the levels of primer type for each application method is shown in Fig. 13.12. This interaction graph was constructed by Minitab. The absence of interaction is evident in the parallelism of the two lines. Furthermore, since a large response indicates greater adhesion force, we conclude that spraying is a superior application method and that primer type 2 is most effective. Therefore, if we wish to operate the process so as to attain maximum adhesion force, we should use primer type 2 and spray all parts.

■ TABLE 13.5
**ANOVA Output from Minitab, Example 13.5**

Two-way ANOVA: Force versus Primer, Method
Analysis of Variance for Force

| Source | DF | SS | MS | F | P |
|---|---|---|---|---|---|
| Primer | 2 | 4.5811 | 2.2906 | 27.86 | 0.000 |
| Method | 1 | 4.9089 | 4.9089 | 59.70 | 0.000 |
| Interaction | 2 | 0.2411 | 0.1206 | 1.47 | 0.269 |
| Error | 12 | 0.9867 | 0.0822 | | |
| Total | 17 | 10.7178 | | | |

Individual 95% CI

| Primer | Mean | ------+---------+---------+---------+-------- |
|---|---|---|
| 1 | 4.78 | (----*----) |
| 2 | 5.68 | (----*----) |
| 3 | 4.50 | (----*----) |
| | | ------+---------+---------+---------+-------- |
| | | 4.50 5.00 5.50 6.00 |

Individual 95% CI

| Method | Mean | ---------+---------+---------+---------+----- |
|---|---|---|
| Dip | 4.467 | (-----*-----) |
| Spray | 5.511 | (-----*-----) |
| | | ---------+---------+---------+---------+----- |
| | | 4.550 4.900 5.250 5.600 |

Confidence intervals on each mean calculated using $MS_E$ as an estimate of $\sigma^2$ and applying the standard confidence interval procedure for the mean of a normal distribution with unknown variance.

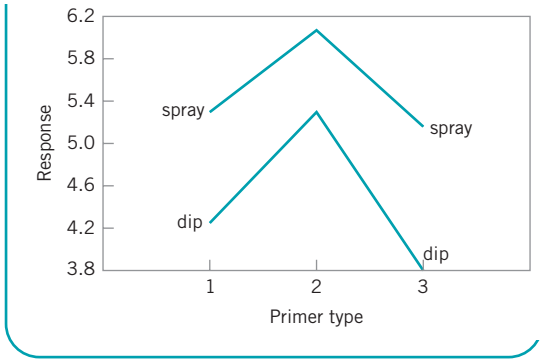



**FIGURE 13.12** Graph of average adhesion force versus primer types for Example 13.5.

### 13.4.3 Residual Analysis

Just as in the single-factor experiments discussed in Chapter 4, the **residuals** from a factorial experiment play an important role in assessing model adequacy. The residuals from a two-factor factorial are

$$e_{ijk} = y_{ijk} - \hat{y}_{ijk}$$
$$= y_{ijk} - \bar{y}_{ij.}$$

That is, the residuals are simply the difference between the observations and the corresponding cell averages.

Table 13.6 presents the residuals for the aircraft primer paint data in Example 13.5. The normal probability plot of these residuals is shown in Fig. 13.13. This plot has tails that do not fall exactly along a straight line passing through the center of the plot, indicating some small problems with the normality assumption, but the departure from normality is not serious. Figures 13.14 and 13.15 plot the residuals versus the levels of primer types and application methods, respectively. There is some indication that primer type 3 results in slightly

**TABLE 13.6**
**Residuals for the Aircraft Primer Paint Experiment**

| Primer Type | Application Method | |
|---|---|---|
| | Dipping | Spraying |
| 1 | –0.26, 0.23, 0.03 | 0.10, –0.40, 0.30 |
| 2 | 0.30, –0.40, 0.10 | –0.26, 0.03, 0.23 |
| 3 | –0.03, –0.13, 0.16 | 0.34, –0.17, –0.17 |

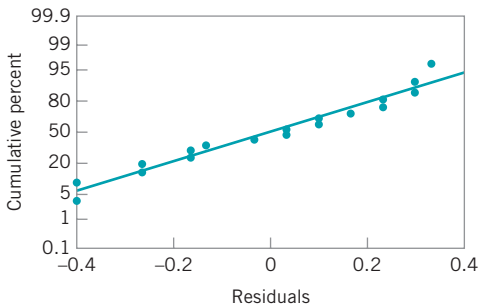



**FIGURE 13.13** Normal probability plot of the residuals from Example 13.5.

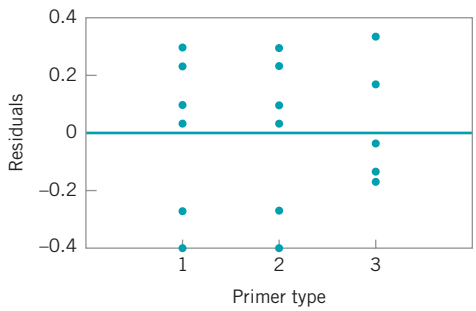
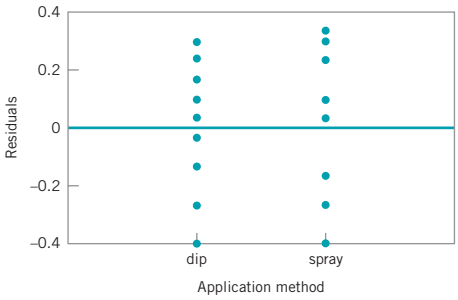
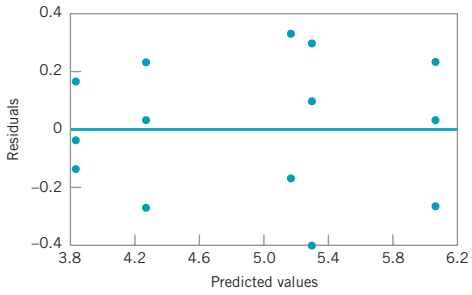



**FIGURE 13.14** Plot of residuals versus primer type.




■ FIGURE 13.15 Plot of residuals versus application method.




■ FIGURE 13.16 Plot of residuals versus predicted values.

lower variability in adhesion force than the other two primers. The graph of residuals versus fitted values in Fig. 13.16 does not reveal any unusual or diagnostic pattern.

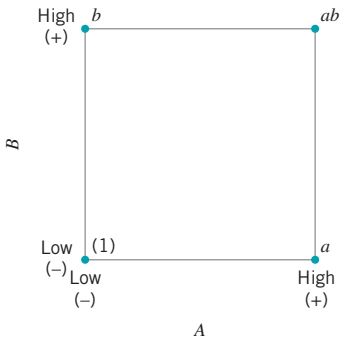
## 13.5 The $2^k$ Factorial Design

Certain special types of factorial designs are very useful in process development and improvement. One of these is a factorial design with $k$ factors, each at two levels. Because each complete replicate of the design has $2^k$ runs, the arrangement is called a **$2^k$ factorial design.** These designs have a greatly simplified analysis, and they also form the basis of many other useful designs.

### 13.5.1 The $2^2$ Design

The simplest type of $2^k$ design is the $2^2$—that is, two factors $A$ and $B$, each at two levels. We usually think of these levels as the "low" or "–" and "high" or "+" levels of the factor. The geometry of the $2^2$ design is shown in Fig. 13.17*a*. Note that the design can be represented geometrically as a square with the $2^2 = 4$ runs forming the corners of the square. Figure 13.17*b* shows the four runs in a tabular format often called the **test matrix** or the **design matrix.** Each run of the test matrix is on the corners of the square and the − and + signs in each row show the settings for the variables $A$ and $B$ for that run.

Another notation is used to represent the runs. In general, a run is represented by a series of lowercase letters. If a letter is present, then the corresponding factor is set at the high level in that run; if it is absent, the factor is run at its low level. For example, run $a$ indicates




(*a*) Design geometry

| A | B |
|---|---|
| – | – |
| + | – |
| – | + |
| + | + |

(*b*) Test matrix

■ FIGURE 13.17 The $2^2$ factorial design.

that factor $A$ is at the high level and factor $B$ is at the low level. The run with both factors at the low level is represented by (1). This notation is used throughout the family of $2^k$ designs. For example, the run in a $2^4$ with $A$ and $C$ at the high level and $B$ and $D$ at the low level is denoted by $ac$.

The effects of interest in the $2^2$ design are the main effects $A$ and $B$ and the two-factor interaction $AB$. Let the letters (1), $a$, $b$, and $ab$ also represent the totals of all $n$ observations taken at these design points. It is easy to estimate the effects of these factors. To estimate the main effect of $A$, we would average the observations on the right side of the square when $A$ is at the high level and subtract from this the average of the observations on the left side of the square where $A$ is at the low level, or

$$\begin{aligned} A &= \bar{y}_{A^+} - \bar{y}_{A^-} \\ &= \frac{a+ab}{2n} - \frac{b+(1)}{2n} \\ &= \frac{1}{2n}\left[a+ab-b-(1)\right] \end{aligned} \tag{13.11}$$

Similarly, the main effect of $B$ is found by averaging the observations on the top of the square where $B$ is at the high level and subtracting the average of the observations on the bottom of the square where $B$ is at the low level:

$$\begin{aligned} B &= \bar{y}_{B^+} - \bar{y}_{B^-} \\ &= \frac{b+ab}{2n} - \frac{a+(1)}{2n} \\ &= \frac{1}{2n}\left[b+ab-a-(1)\right] \end{aligned} \tag{13.12}$$

Finally, the $AB$ interaction is estimated by taking the difference in the diagonal averages in Fig. 13.17, or

$$\begin{aligned} AB &= \frac{ab+(1)}{2n} - \frac{a+b}{2n} \\ &= \frac{1}{2n}\left[ab+(1)-a-b\right] \end{aligned} \tag{13.13}$$

The quantities in brackets in equations (13.11), (13.12), and (13.13) are called **contrasts.** For example, the $A$ contrast is

$$\text{Contrast}_A = a+ab-b-(1)$$

In these equations, the contrast coefficients are always either + 1 or − 1. A table of plus and minus signs, such as Table 13.7, can be used to determine the sign on each run for a particular

■ TABLE 13.7
**Signs for Effects in the $2^2$ Design**

| | | Factorial Effect | | | |
|---|---|---|---|---|---|
| **Run** | | ***I*** | ***A*** | ***B*** | ***AB*** |
| 1 | (1) | + | − | − | + |
| 2 | *a* | + | + | − | − |
| 3 | *b* | + | − | + | − |
| 4 | *ab* | + | + | + | + |

contrast. The column headings for the table are the main effects $A$ and $B$, the $AB$ interaction, and $I$, which represents the total. The row headings are the runs. Note that the signs in the $AB$ column are the product of signs from columns $A$ and $B$. To generate a contrast from this table, multiply the signs in the appropriate column of Table 13.7 by the runs listed in the rows and add.

To obtain the sums of squares for $A$, $B$, and $AB$, we use the following result.

$$SS = \frac{(\text{contrast})^2}{n\sum(\text{contrast coefficients})^2} \tag{13.14}$$

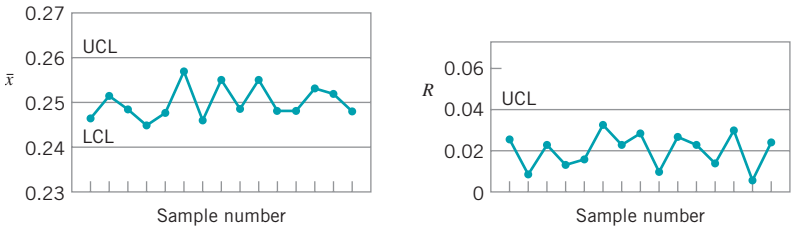
Therefore, the sums of squares for $A$, $B$, and $AB$ are

$$SS_A = \frac{[a + ab - b - (1)]^2}{4n}$$
$$SS_B = \frac{[b + ab - a - (1)]^2}{4n} \tag{13.15}$$
$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{4n}$$

The analysis of variance is completed by computing the total sum of squares $SS_T$ (with $4n - 1$ degrees of freedom) as usual, and obtaining the error sum of squares $SS_E$ [with $4(n - 1)$ degrees of freedom] by subtraction.

## EXAMPLE 13.6 The Router Experiment

A router is used to cut registration notches in printed circuit boards. The average notch dimension is satisfactory, and the process is in statistical control (see the $\bar{x}$ and $R$ control charts in Fig. 13.18), but there is too much variability in the process. This excess variability leads to problems in board assembly. The components are inserted into the board using automatic equipment, and the variability in notch dimension causes improper board registration. As a result, the auto-insertion equipment does not work properly. How would you improve this process?




■ **FIGURE 13.18** $\bar{x}$ and $R$ control charts on notch dimension, Example 13.6.

## SOLUTION

Since the process is in statistical control, the quality improvement team assigned to this project decided to use a designed experiment to study the process. The team considered two factors: bit size ($A$) and speed ($B$). Two levels were chosen for each factor (bit size $A$ at $\frac{1}{16}''$ and $\frac{1}{8}''$ and speed $B$ at 40 rpm and 80 rpm), and a $2^2$ design was set up. Since variation in notch dimension was difficult to measure directly, the team decided to measure it indirectly. Sixteen test boards were instrumented with accelerometers that allowed vibration on the ($X$, $Y$, $Z$) coordinate axes to be measured. The resultant vector of these three components was used as the response variable. Since vibration at the surface of the board when it is cut is directly related to variability in notch dimension, reducing vibration levels will also reduce the variability in notch dimension.

Four boards were tested at each of the four runs in the experiment, and the resulting data are shown in Table 13-8.

■ **TABLE 13.8**
**Data from the Router Experiment**

| Run | | Factors $A$ | Factors $B$ | Vibration | | | | Total |
|---|---|---|---|---|---|---|---|---|
| 1 | (1) | − | − | 18.2 | 18.9 | 12.9 | 14.4 | 64.4 |
| 2 | $a$ | + | − | 27.2 | 24.0 | 22.4 | 22.5 | 96.1 |
| 3 | $b$ | − | + | 15.9 | 14.5 | 15.1 | 14.2 | 59.7 |
| 4 | $ab$ | + | + | 41.0 | 43.9 | 36.3 | 39.9 | 161.1 |

Using equations (13.11), (13.12), and (13.13), we can compute the factor effect estimates as follows:

$$A = \frac{1}{2n}\left[a + ab - b - (1)\right]$$
$$= \frac{1}{2(4)}\left[96.1 + 161.1 - 59.7 - 64.4\right] = \frac{133.1}{8} = 16.64$$
$$B = \frac{1}{2n}\left[b + ab - a - (1)\right]$$
$$= \frac{1}{2(4)}\left[59.7 + 161.1 - 96.1 - 64.4\right] = \frac{60.3}{8} = 7.54$$
$$AB = \frac{1}{2n}\left[ab + (1) - a - b\right]$$
$$= \frac{1}{2(4)}\left[161.1 + 64.4 - 96.1 - 59.7\right] = \frac{69.7}{8} = 8.71$$

All the numerical effect estimates seem large. For example, when we change factor $A$ from the low level to the high level (bit size from $\frac{1}{16}''$ to $\frac{1}{8}''$), the average vibration level increases by 16.64 cps.

The magnitude of these effects may be confirmed with the analysis of variance, which is summarized in Table 13.9. The sums of squares in this table for main effects and interaction were computed using equation (13.15). The analysis of variance confirms our conclusions that were obtained by initially examining the magnitude and direction of the factor effects; both bit size and speed are important, and there is interaction between two variables.

■ **TABLE 13.9**
**Analysis of Variance for the Router Experiment**

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Square | $F_0$ | $P$-value |
|---|---|---|---|---|---|
| Bit size ($A$) | 1107.226 | 1 | 1107.226 | 185.25 | $1.17 \times 10^{-8}$ |
| Speed ($B$) | 227.256 | 1 | 227.256 | 38.03 | $4.82 \times 10^{-5}$ |
| $AB$ | 303.631 | 1 | 303.631 | 50.80 | $1.20 \times 10^{-5}$ |
| Error | 71.723 | 12 | 5.977 | | |
| Total | 1709.836 | 15 | | | |

***Regression Model and Residual Analysis.*** It is easy to obtain the residuals from a $2^k$ design by fitting a **regression model** to the data. For the router experiment, the regression model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon$$

where the factors $A$ and $B$ are represented by coded variables $x_1$ and $x_2$, and the $AB$ interaction is represented by the cross-product term in the model, $x_1 x_2$. The low and high levels of each factor are assigned the values $x_j = -1$ and $x_j = +1$, respectively. The coefficients $\beta_0$, $\beta_1$, $\beta_2$, and $\beta_{12}$ are called **regression coefficients,** and $\varepsilon$ is a random error term, similar to the error term in an analysis of variance model.

The fitted regression model is

$$\hat{y} = 23.83 + \left(\frac{16.64}{2}\right)x_1 + \left(\frac{7.54}{2}\right)x_2 + \left(\frac{8.71}{2}\right)x_1 x_2$$

where the estimate of the intercept $\hat{\beta}_0$ is the grand average of all 16 observations ($\bar{y}$) and the estimates of the other regression coefficients $\hat{\beta}_j$ are one-half the effect estimate for the corresponding factor. [Each regression coefficient estimate is one-half the effect estimate because regression coefficients measure the effect of a unit change in $x_j$ on the mean of $y$, and the effect estimate is based on a two-unit change (from −1 to +1).]

This model can be used to obtain the predicted values of vibration level at any point in the region of experimentation, including the four points in the design. For example, consider the point with the small bit ($x_1 = -1$) and low speed ($x_2 = -1$). The predicted vibration level is

$$\hat{y} = 23.83 + \left(\frac{16.64}{2}\right)(-1) + \left(\frac{7.54}{2}\right)(-1) + \left(\frac{8.71}{2}\right)(-1)(-1) = 16.1$$

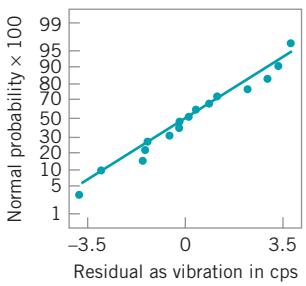
The four residuals corresponding to the observations at this design point are found by taking the difference between the actual observation and the predicted value as follows:

$$e_1 = 18.2 - 16.1 = 2.1 \quad e_3 = 12.9 - 16.1 = -3.2$$
$$e_2 = 18.9 - 16.1 = 2.8 \quad e_4 = 14.4 - 16.1 = -1.7$$

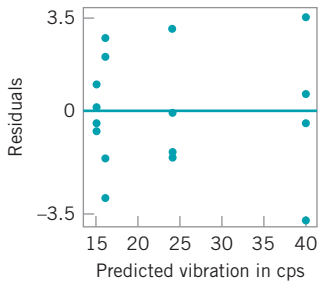
The residuals at the other three runs would be computed similarly.

Figures 13.19 and 13.20 present the normal probability plot and the plot of residuals versus the fitted values, respectively. The normal probability plot is satisfactory, as is the plot of residuals versus $\hat{y}$, although this latter plot does give some indication that there may be less variability in the data at the point of lowest predicted vibration level.

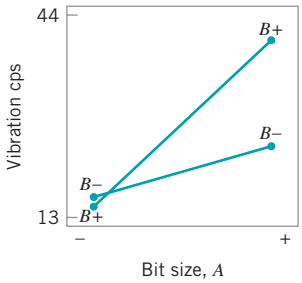
***Practical Interpretation of Example 13.6.*** Since both factors $A$ (bit size) and $B$ (speed) have large, positive effects, we could reduce vibration levels by running both factors



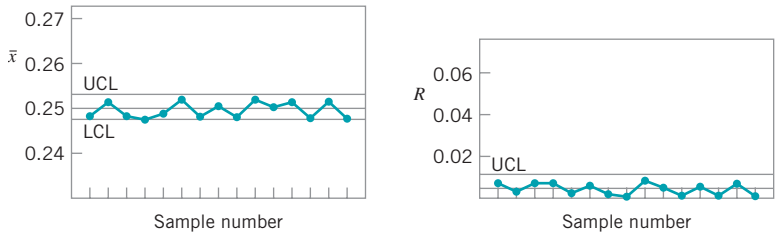

■ **FIGURE 13.19** Normal probability plot, Example 13.6.




■ **FIGURE 13.20** Plot of residuals versus $\hat{y}$, Example 13.6.




■ FIGURE 13.21 *AB* interaction plot.




■ FIGURE 13.22 $\bar{x}$ and $R$ charts for the router process after the experiment.

at the low level. However, with both bit size and speed at low level, the production rate could be unacceptably low. The *AB* interaction provides a solution to this potential dilemma. Figure 13.21 presents the two-factor *AB* interaction plot. Note that the large positive effect of speed occurs primarily when bit size is at the high level. If we use the small bit, then either speed level will provide lower vibration levels. If we run with speed high and use the small bit, the production rate will be satisfactory.

When manufacturing implemented this set of operating conditions, the result was a dramatic reduction in variability in the registration notch dimension. The process remained in statistical control, as the control charts in Fig. 13.22 imply, and the reduced variability dramatically improved the performance of the auto-insertion process.

***Analysis Procedure for Factorial Experiments.*** Table 13.10 summarizes the sequence of steps that is usually employed to analyze factorial experiments. These steps were followed in the analysis of the router experiment in Example 13.6. Recall that our first activity, after the experiment was run, was to estimate the effect of the factors bit size, speed, and the two-factor interaction. The preliminary model that we used in the analysis was the two-factor factorial model with interaction. Generally, in any factorial experiment with replication, we will almost always use the full factorial model as the preliminary model. We tested for significance of factor effects by using the analysis of variance. Since the residual analysis was satisfactory, and both main effects and the interaction term were significant, there was no need to refine the model. Therefore, we were able to interpret the results in terms of the original full factorial model, using the two-factor interaction graph in Fig. 13.21. Sometimes refining the model includes deleting terms from the final model that are not significant, or taking other actions that may be indicated from the residual analysis.

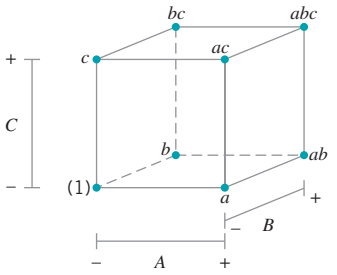
Several statistics software packages include special routines for the analysis of two-level factorial designs. Many of these packages follow an analysis process similar to the one we have outlined. We will illustrate this analysis procedure again several times in this chapter.

### 13.5.2 The $2^k$ Design for $k \geq 3$ Factors

The methods presented in the previous section for factorial designs with $k = 2$ factors each at two levels can be easily extended to more than two factors. For example, consider $k = 3$

■ TABLE 13.10
**Analysis Procedure for Factorial Designs**

| | |
|---|---|
| 1. Estimate the factor effects | 4. Analyze residuals |
| 2. Form preliminary model | 5. Refine model, if necessary |
| 3. Test for significance of factor effects | 6. Interpret results |




(a) Design geometry

| A | B | C |
|---|---|---|
| − | − | − |
| + | − | − |
| − | + | − |
| + | + | − |
| − | − | + |
| + | − | + |
| − | + | + |
| + | + | + |

(b) Test matrix

■ **FIGURE 13.23** The $2^3$ factorial design.

factors, each at two levels. This design is a $2^3$ factorial design, and it has eight factor-level combinations. Geometrically, the design is a cube as shown in Fig. 13.23*a*, with the eight runs forming the corners of the cube. Figure 13.23*b* shows the **test design matrix.** This design allows three main effects to be estimated (*A*, *B*, and *C*) along with three two-factor interactions (*AB*, *AC*, and *BC*) and a three-factor interaction (*ABC*). Thus, the full factorial model could be written symbolically as

$$y = \mu + A + B + C + AB + AC + BC + ABC + \varepsilon$$

where $\mu$ is an overall mean, $\varepsilon$ is a random error term assumed to be NID(0, $\sigma^2$), and the uppercase letters represent the main effects and interactions of the factors [note that we could have used Greek letters for the main effects and interactions, as in equation (13.2)].

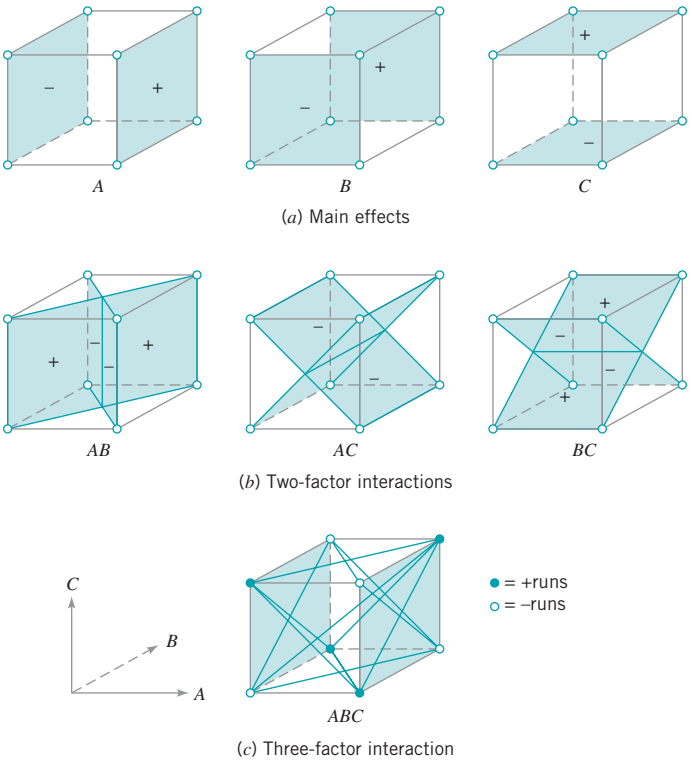
The main effects can be estimated easily. Remember that the lowercase letters (1), *a*, *b*, *ab*, *c*, *ac*, *bc*, and *abc* represent the total of all *n* replicates at each of the eight runs in the design. Referring to the cube in Fig. 13.23, we would estimate the main effect of *A* by averaging the four runs on the right side of the cube where *A* is at the high level and subtracting from that quantity the average of the four runs on the left side of the cube where *A* is at the low level. This gives

$$A = \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{1}{4n}\left[a + ab + ac + abc - b - c - bc - (1)\right] \qquad (13.16)$$

In a similar manner, the effect of *B* is the average difference of the four runs in the back face of the cube and the four in the front, or

$$B = \bar{y}_{B^+} - \bar{y}_{B^-} = \frac{1}{4n}\left[b + ab + bc + abc - a - c - ac - (1)\right] \qquad (13.17)$$

and the effect of *C* is the average difference between the four runs in the top face of the cube and the four in the bottom, or




■ **FIGURE 13.24** Geometric presentation of contrasts corresponding to the main effects and interaction in the $2^3$ design.

$$C = \bar{y}_{C^+} - \bar{y}_{C^-} = \frac{1}{4n}\left[c + ac + bc + abc - a - b - ab - (1)\right] \tag{13.18}$$

The top row of Fig. 13.24 shows how the main effects of the three factors are computed.

Now consider the two-factor interaction $AB$. When $C$ is at the low level, $AB$ is simply the average difference in the $A$ effect at the two levels of $B$, or

$$AB(C \text{ low}) = \frac{1}{2n}[ab - b] - \frac{1}{2n}[a - (1)]$$

Similarly, when $C$ is at the high level, the $AB$ interaction is

$$AB(C \text{ high}) = \frac{1}{2n}[abc - bc] - \frac{1}{2n}[ac - c]$$

The $AB$ interaction is the average of these two components, or

$$AB = \frac{1}{4n}\left[ab + (1) + abc + c - b - a - bc - ac\right] \tag{13.19}$$

Note that the *AB* interaction is simply the difference in averages on two diagonal planes in the cube (refer to the left-most cube in the middle row of Fig. 13.24).

Using a similar approach, we see from the middle row of Fig. 13.24 that the *AC* and *BC* interaction effect estimates are as follows:

$$AC = \frac{1}{4n}\left[ac + (1) + abc + b - a - c - ab - bc\right] \tag{13.20}$$

$$BC = \frac{1}{4n}\left[bc + (1) + abc + a - b - c - ab - ac\right] \tag{13.21}$$

The *ABC* interaction effect is the average difference between the *AB* interaction at the two levels of *C*. Thus

$$ABC = \frac{1}{4n}\left\{\left[abc - bc\right] - \left[ac - c\right] - \left[ab - b\right] + \left[a - (1)\right]\right\}$$

or

$$ABC = \frac{1}{4n}\left[abc - bc - ac + c - ab + b + a - (1)\right] \tag{13.22}$$

This effect estimate is illustrated in the bottom row of Fig. 13.24.

The quantities in brackets in equations (13.16) through (13.22) are contrasts in the eight factor-level combinations. These contrasts can be obtained from a table of plus and minus signs for the $2^3$ design, shown in Table 13.11. Signs for the main effects (columns *A*, *B*, and *C*) are obtained by associating a plus with the high level and a minus with the low level. Once the signs for the main effects have been established, the signs for the remaining columns are found by multiplying the appropriate preceding columns, row by row. For example, the signs in column *AB* are the product of the signs in columns *A* and *B*.

■ **TABLE 13.11**
**Signs for Effects in the $2^3$ Design**

| Treatment Combination | Factorial Effect | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | *I* | *A* | *B* | *AB* | *C* | *AC* | *BC* | *ABC* |
| (1) | + | − | − | + | − | + | + | − |
| *a* | + | + | − | − | − | − | + | + |
| *b* | + | − | + | − | − | + | − | + |
| *ab* | + | + | + | + | − | − | − | − |
| *c* | + | − | − | + | + | − | − | + |
| *ac* | + | + | − | − | + | + | − | − |
| *bc* | + | − | + | − | + | − | + | − |
| *abc* | + | + | + | + | + | + | + | + |

Table 13.11 has several interesting properties:

1. Except for the identity column $I$, each column has an equal number of plus and minus signs.
2. The sum of products of signs in any two columns is zero; that is, the columns in the table are *orthogonal*.
3. Multiplying any column by column $I$ leaves the column unchanged; that is, $I$ is an *identity element*.
4. The product of any two columns yields a column in the table; for example, $A \times B = AB$, and $AB \times ABC = A^2B^2C = C$, since any column multiplied by itself is the identity column.

The estimate of any main effect or interaction is determined by multiplying the factor-level combinations in the first column of the table by the signs in the corresponding main effect or interaction column, adding the result to produce a contrast, and then dividing the contrast by one-half the total number of runs in the experiment. Expressed mathematically,

$$\text{Effect} = \frac{\text{Contrast}}{n2^{k-1}} \tag{13.23}$$

The sum of squares for any effect is

$$SS = \frac{(\text{Contrast})^2}{n2^k} \tag{13.24}$$

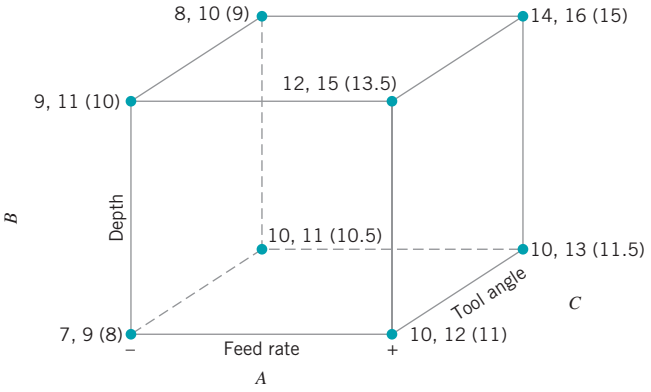
## EXAMPLE 13.7 A $2^3$ Factorial Design

An experiment was performed to investigate the surface finish of a metal part. The experiment is a $2^3$ factorial design in the factors feed rate ($A$), depth of cut ($B$), and tool angle ($C$), with $n = 2$ replicates. Table 13.12 presents the observed surface-finish data for this experiment, and the design is shown graphically in Fig. 13.25. Analyze and interpret the data from this experiment.

**TABLE 13.12**
**Surface-Finish Data for Example 13.7**

| | | Design Factors | | | | |
|---|---|---|---|---|---|---|
| **Run** | | ***A*** | ***B*** | ***C*** | **Surface Finish** | **Totals** |
| 1 | (1) | −1 | −1 | −1 | 9, 7 | 16 |
| 2 | *a* | 1 | −1 | −1 | 10, 12 | 22 |
| 3 | *b* | −1 | 1 | −1 | 9, 11 | 20 |
| 4 | *ab* | 1 | 1 | −1 | 12, 15 | 27 |
| 5 | *c* | −1 | −1 | 1 | 11, 10 | 21 |
| 6 | *ac* | 1 | −1 | 1 | 10, 13 | 23 |
| 7 | *bc* | −1 | 1 | 1 | 10, 8 | 18 |
| 8 | *abc* | 1 | 1 | 1 | 16, 14 | 30 |

(*continued*)




■ **FIGURE 13.25** $2^3$ design for the surface-finish experiment in Example 13.7 (the numbers in parentheses are the average responses at each design point).

## SOLUTION

The main effects may be estimated using equations (13.16) through (13.22). The effect of $A$, for example, is

$$A = \frac{1}{4n}[a + ab + ac + abc - b - c - bc - (1)]$$
$$= \frac{1}{4(2)}[22 + 27 + 23 + 30 - 20 - 21 - 18 - 16]$$
$$= \frac{1}{8}[27] = 3.375$$

and the sum of squares for $A$ is found using equation (13.24):

$$SS_A = \frac{(\text{Contrast}_A)^2}{n2^k}$$
$$= \frac{(27)^2}{2(8)} = 45.5625$$

It is easy to verify that the other effect estimates and sums of squares are

$$B = 1.625 \qquad SS_B = 10.5625$$
$$C = 0.875 \qquad SS_C = 3.0625$$
$$AB = 1.375 \qquad SS_{AB} = 7.5625$$
$$AC = 0.125 \qquad SS_{AC} = 0.0625$$
$$BC = -0.625 \qquad SS_{BC} = 1.5625$$
$$ABC = 1.125 \qquad SS_{ABC} = 5.0625$$

From examining the magnitude of the effects, feed rate (factor $A$) is clearly dominant, followed by depth of cut ($B$) and the $AB$ interaction, although the interaction effect is relatively small. The analysis of variance for the full factorial model is summarized in Table 13.13. Based on the $P$-values, it is clear that the feed rate ($A$) is highly significant.

■ **TABLE 13.13**
**Analysis of Variance for the Surface-Finish Experiment**

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Square | $F_0$ | $P$-value |
|---|---|---|---|---|---|
| $A$ | 45.5625 | 1 | 45.5625 | 18.69 | $2.54 \times 10^{-3}$ |
| $B$ | 10.5625 | 1 | 10.5625 | 4.33 | 0.07 |
| $C$ | 3.0625 | 1 | 3.0625 | 1.26 | 0.29 |
| $AB$ | 7.5625 | 1 | 7.5625 | 3.10 | 0.12 |
| $AC$ | 0.0625 | 1 | 0.0625 | 0.03 | 0.88 |
| $BC$ | 1.5625 | 1 | 1.5625 | 0.64 | 0.45 |
| $ABC$ | 5.0625 | 1 | 5.0625 | 2.08 | 0.19 |
| Error | 19.5000 | 8 | 2.4375 | | |
| Total | 92.9375 | 15 | | | |

**TABLE 13.14**
**Analysis of Variance from Minitab for the Surface-Finish Experiment**

Factorial Design

Full Factorial Design

| | | | |
|---|---|---|---|
| Factors: | 3 | Base Design: | 3, 8 |
| Runs: | 16 | Replicates: | 2 |
| Blocks: | none | Center pts (total): | 0 |

All terms are free from aliasing

Fractional Factorial Fit: Finish versus A, B, C

Estimated Effects and Coefficients for Finish (coded units)

| Term | Effect | Coef | SE Coef | T | P |
|---|---|---|---|---|---|
| Constant | | 11.0625 | 0.3903 | 28.34 | 0.000 |
| A | 3.3750 | 1.6875 | 0.3903 | 4.32 | 0.003 |
| B | 1.6250 | 0.8125 | 0.3903 | 2.08 | 0.071 |
| C | 0.8750 | 0.4375 | 0.3903 | 1.12 | 0.295 |
| A*B | 1.3750 | 0.6875 | 0.3903 | 1.76 | 0.116 |
| A*C | 0.1250 | 0.0625 | 0.3903 | 0.16 | 0.877 |
| B*C | –0.6250 | –0.3125 | 0.3903 | –0.80 | 0.446 |
| A*B*C | 1.1250 | 0.5625 | 0.3903 | 1.44 | 0.188 |

Analysis of Variance for Finish (coded units)

| Source | DF | Seq SS | Adj SS | Adj MS | F | P |
|---|---|---|---|---|---|---|
| Main Effects | 3 | 59.187 | 59.187 | 19.729 | 8.09 | 0.008 |
| 2-Way Interactions | 3 | 9.187 | 9.187 | 3.062 | 1.26 | 0.352 |
| 3-Way Interactions | 1 | 5.062 | 5.062 | 5.062 | 2.08 | 0.188 |
| Residual Error | 8 | 19.500 | 19.500 | 2.437 | | |
| Pure Error | 8 | 19.500 | 19.500 | 2.438 | | |
| Total | 15 | 92.937 | | | | |

Many computer programs analyze the $2^k$ factorial design. Table 13.14 is the output from Minitab. Although at first glance the two tables seem somewhat different, they actually provide the same information. The analysis of variance displayed in the lower portion of Table 13.14 presents $F$-ratios computed on important groups of model terms; main effects, two-way interactions, and the three-way interaction. The mean square for each group of model terms was obtained by combining the sums of squares for each model component and dividing by the number of degrees of freedom associated with that group of model terms.

A $t$-test is used to test the significance of each individual term in the model. These $t$-tests are shown in the upper portion of Table 13.14. Note that a "coefficient estimate" is given for each variable in the full factorial model. These are actually the estimates of the coefficients in the regression model that would be used to predict surface finish in terms of the variables in the full factorial model. Each $t$-value is computed according to

$$t_0 = \frac{\hat{\beta}}{s.e.\left(\hat{\beta}\right)}$$

where $\hat{\beta}$ is the coefficient estimate and $s.e.\,(\hat{\beta})$ is the estimated standard error of the coefficient. For a $2^k$ factorial design, the estimated standard error of the coefficient is

$$s.e.\left(\hat{\beta}\right) = \sqrt{\frac{\hat{\sigma}^2}{n2^k}}$$

We use the error or residual mean square from the analysis of variance as the estimate $\hat{\sigma}^2$. In our example,

$$s.e.\left(\hat{\beta}\right) = \sqrt{\frac{2.4375}{2\left(2^3\right)}} = 0.390312$$

as shown in Table 13.14. It is easy to verify that dividing any coefficient estimate by its estimated standard error produces the $t$-value for testing whether the corresponding regression coefficient is zero.

The $t$-tests in Table 13.14 are equivalent to the ANOVA $F$-tests in Table 13.13. You may have suspected this already, since the $P$-values in the two tables are identical to two decimal places. Furthermore, note that the square of any $t$-value in Table 13.14 produces the corresponding $F$-ratio value in Table 13.13.

(*continued*)

In general, the square of a $t$ random variable with $v$ degrees of freedom results in an $F$ random variable with one numerator degree of freedom and $v$ denominator degrees of freedom. This explains the equivalence of the two procedures used to conduct the analysis of variance for the surface-finish experiment data.

Based on the ANOVA results, we conclude that the full factorial model in all these factors is unnecessary, and that a reduced model including fewer variables is more appropriate. The main effects of $A$ and $B$ both have relatively small $P$-values ($< 0.10$), and this $AB$ interaction is the next most important effect ($P$-value $\cong 0.12$). The regression model that we would use to represent this process is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon$$

where $x_1$ represents factor $A$, $x_2$ represents factor $B$, and $x_1x_2$ represents the $AB$ interaction. The regression coefficients $\hat{\beta}_1$, $\hat{\beta}_2$, and $\hat{\beta}_{12}$ are one-half the corresponding effect estimates and $\hat{\beta}_0$ is the grand average. Thus

$$\hat{y} = 11.0625 + \left(\frac{3.375}{2}\right)x_1 + \left(\frac{1.625}{2}\right)x_2 + \left(\frac{1.375}{2}\right)x_1x_2$$
$$= 11.0625 + 1.6875x_1 + 0.8125x_2 + 0.6875x_1x_2$$

Note that we can read the values of $\hat{\beta}_0$, $\hat{\beta}_1$, $\hat{\beta}_2$, and $\hat{\beta}_{12}$ directly from the "coefficient" column of Table 13.14.

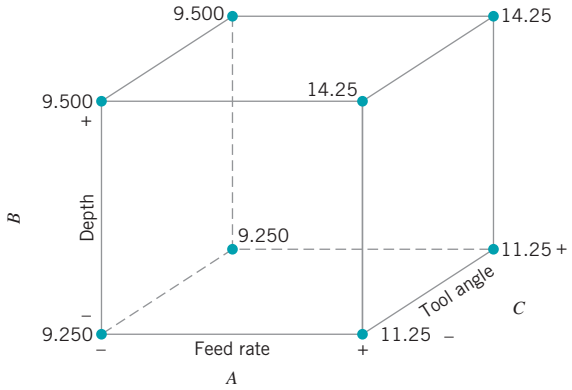
This regression model can be used to predict surface finish at any point in the original experimental region. For example, consider the point where all three variables are at the low level. At this point, $x_1 = x_2 = -1$, and the predicted value is

$$\hat{y} = 11.0625 + 1.6875(-1) + 0.8125(-1) + 0.6875(-1)(-1) = 9.25$$

Figure 13.26 shows the predicted values at each point in the original experimental design.

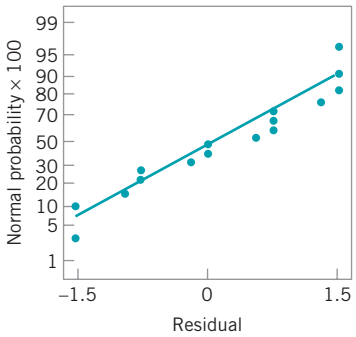
The residuals can be obtained as the difference between the observed and predicted values of surface finish at each design point. For the point where all three factors $A$, $B$, and $C$ are at the low level, the observed values of surface finish are 9 and 7, so the residuals are $9 - 9.25 = -0.25$ and $7 - 9.25 = -2.25$.

A normal probability plot of the residuals is shown in Fig. 13.27. Since the residuals lie approximately along a straight line, we do not suspect any severe nonnormality in the data. There are also no indications of outliers. It would also be helpful to plot the residuals versus the predicted values and against each of the factors $A$, $B$, and $C$. These plots do not indicate any potential model problems.




■ **FIGURE 13.26** Predicted values of surface finish at each point in the original design, Example 13.7.

Finally, we can provide a practical interpretation of the results of our experiment. Both main effects $A$ and $B$ are positive, and since small values of the surface finish response are desirable, this would suggest that both $A$ (feed rate) and $B$ (depth of cut) should be run at the low level. However, the model has an interaction term, and the effect of this interaction should be taken into account when drawing conclusions. We could do this by examining an interaction plot, as in Example 13.6 (see Fig. 13.21). Alternatively, the cube plot of predicted responses in Fig. 13.26 can also be used for model interpretation. This figure indicates that the lowest values of predicted surface finish will be obtained when $A$ and $B$ are at the low level.




■ **FIGURE 13.27** Normal probability plot of residuals, Example 13.7.

***Some Comments on the Regression Model.*** In the two previous examples, we used a **regression model** to summarize the results of the experiment. In general, a regression model is an equation of the form

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_k x_k + \varepsilon \tag{13.25}$$

where $y$ is the response variable, the $x$'s are a set of regressor or predictor variables, the $\beta$'s are the regression coefficients, and $\varepsilon$ is an error term, assumed to be NID(0, $\sigma^2$). In our examples,

we had $k = 2$ factors and the models had an interaction term, so the specific form of the regression model that we fit was

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon$$

In general, the regression coefficients in these models are estimated using the **method of least squares;** that is, the $\hat{\beta}$'s are chosen so as to minimize the sum of the squares of the errors (the $\varepsilon$'s). Refer to Chapter 4 for an introduction to least squares regression. However, in the special case of a $2^k$ design, it is extremely easy to find the least squares estimates of the $\beta$'s. The least squares estimate of any regression coefficient $\beta$ is simply one-half of the corresponding factor effect estimate. Recall that we have used this result to obtain the regression models in Examples 13.6 and 13.7. Also, please remember that this result only works for a $2^k$ factorial design, and it assumes that the $x$'s are coded variables over the range $-1 \le x \le +1$ that represent the design factors.

It is very useful to express the results of a designed experiment in terms of a **model,** which will be a valuable aid in interpreting the experiment. Recall that we used the cube plot of predicted values from the model in Fig. 13.27 to find appropriate settings for feed rate and depth of cut in Example 13.7. More general graphical displays can also be useful. For example, consider the model for surface finish in terms of feed rate ($x_1$) and depth of cut ($x_2$) without the interaction term

$$\hat{y} = 11.0625 + 1.6875 x_1 + 0.8125 x_2$$

Note that the model was obtained simply by deleting the interaction term from the original model. This can only be done if the variables in the experimental design are **orthogonal,** as they are in a $2^k$ design. Figure 13.28 plots the predicted value of surface finish ($\hat{y}$) in terms of the two process variables $x_1$ and $x_2$. Figure 13.28*a* is a three-dimensional plot showing the plane of predicted response values generated by the regression model. This type of display is called a **response surface plot,** and the regression model used to generate the graph is often called a **first-order response surface model.** The graph in Fig. 13.28*b* is a two-dimensional **contour plot** obtained by looking down on the three-dimensional response surface plot and connecting points of constant surface finish (response) in the $x_1$–$x_2$ plane. The lines of constant response are straight lines because the response surface is first-order; that is, it contains only the main effects $x_1$ and $x_2$.

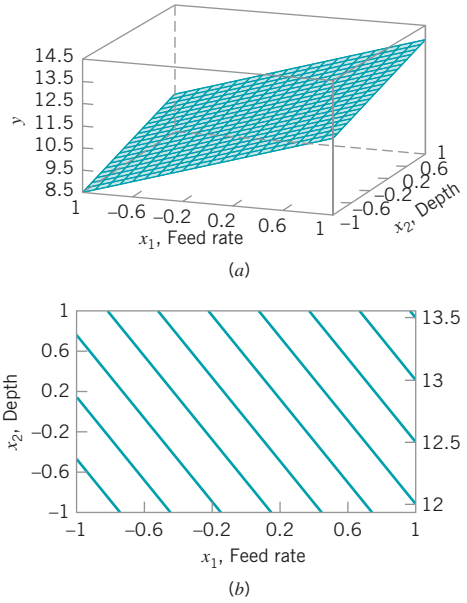
In Example 13.7, we actually fit a first-order model with interaction:

$$\hat{y} = 11.0625 + 1.6875 x_1 + 0.8125 x_2 + 0.6875 x_1 x_2$$

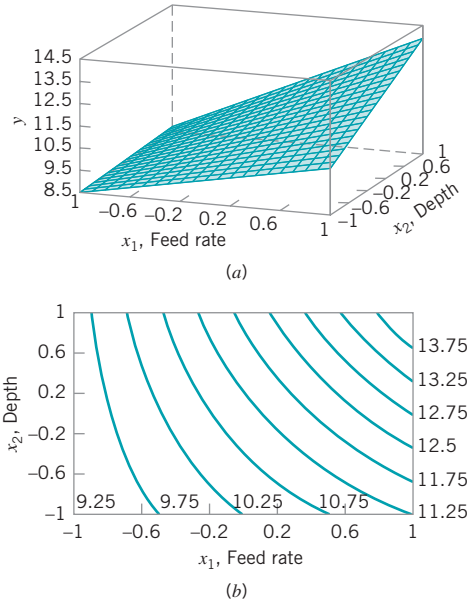
Figure 13.29*a* is the three-dimensional response surface plot for this model and Fig. 13.29*b* is the contour plot. Note that the effect of adding the interaction term to the model is to introduce **curvature** into the response surface; in effect, the plane is "twisted" by the interaction effect.

Inspection of a response surface makes interpretation of the results of an experiment very simple. For example, note from Fig. 13.29 that if we wish to minimize the surface-finish response, we need to run $x_1$ and $x_2$ at (or near) their low levels. We reached the same conclusion by inspection of the cube plot in Fig. 13.26. However, suppose we needed to obtain a particular value of surface finish, say 10.25 (the surface might need to be this rough so that a coating will adhere properly). Figure 13.29*b* indicates that there are many combinations of $x_1$ and $x_2$ that will allow the process to operate on the contour line $\hat{y} = 10.25$. The experimenter might select a set of operating conditions that maximized $x_1$ subject to $x_1$ and $x_2$ giving a predicted response on or near to the contour $\hat{y} = 10.25$, as this would satisfy the surface-finish objective while simultaneously making the feed rate as large as possible, which would maximize the production rate.

Response surface models have many uses. In Chapter 14 we will give an overview of some aspects of response surfaces and how they can be used for process improvement and optimization. However, note how useful the response surface was, even in this simple example. This is why we tell experimenters that **the objective of every designed experiment is a quantitative model of the process.**




■ **FIGURE 13.28** (*a*) Response surface for the model $\hat{y} = 11.0625 + 1.6875x_1 + 0.8125x_2$. (*b*) The contour plot.




■ **FIGURE 13.29** (*a*) Response surface for the model $\hat{y} = 11.0625 + 1.6875x_1 + 0.8125x_2 + 0.6875x_1x_2$. (*b*) The contour plot.

***Projection of $2^k$ Designs.*** Any $2^k$ design will collapse or project into another two-level factorial design in fewer variables if one or more of the original factors are dropped. Usually this will provide additional insight into the remaining factors. For example, consider the surface-finish experiment. Since factor $C$ and all its interactions are negligible, we could eliminate factor $C$ from the design. The result is to collapse the cube in Fig. 13.25 into a square in the $A$-$B$ plane; however, each of the four runs in the new design has four replicates. In general, if we delete $h$ factors so that $r = k - h$ factors remain, the original $2^k$ design with $n$ replicates will project into a $2^r$ design with $n2^h$ replicates.

***Other Methods for Judging the Significance of Effects.*** The analysis of variance is a formal way to determine which effects are nonzero. Two other methods are useful. In the first method, we can calculate the standard errors of the effects and compare the magnitude of the effects to their standard errors. The second method uses normal probability plots to assess the importance of the effects.

The standard error of any effect estimate in a $2^k$ design is given by

$$s.e.\ (\text{Effect}) = \sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}} \tag{13.26}$$

where $\hat{\sigma}^2$ is an estimate of the experimental error variance $\sigma^2$. We usually take the error (or residual) mean square from the analysis of variance as the estimate $\hat{\sigma}^2$ of $\sigma^2$.

As an illustration for the surface-finish experiment, we find that $MS_E = \hat{\sigma}^2 = 2.4375$, and the standard error of each effect is

$$\begin{aligned} s.e.\ (\text{Effect}) &= \sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}} \\ &= \sqrt{\frac{2.4375}{(2)2^{3-2}}} \\ &= 0.78 \end{aligned}$$

Therefore, two standard deviation limits on the effect estimates are

$$\begin{aligned} A&: \ 3.375 \pm 1.56 \\ B&: \ 1.625 \pm 1.56 \\ C&: \ 0.875 \pm 1.56 \\ AB&: \ 1.375 \pm 1.56 \\ AC&: \ 0.125 \pm 1.56 \\ BC&: -0.625 \pm 1.56 \\ ABC&: \ 1.125 \pm 1.56 \end{aligned}$$

These intervals are approximate 95% confidence intervals. They indicate that the two main effects $A$ and $B$ are important but that the other effects are not, since the intervals for all effects except $A$ and $B$ include zero. These conclusions are similar to those found in Example 13.7.

Normal probability plots can also be used to judge the significance of effects. We will illustrate that method in the next section.

### 13.5.3 A Single Replicate of the $2^k$ Design

As the number of factors in a factorial experiment grows, the number of effects that can be estimated also grows. For example, a $2^4$ experiment has 4 main effects, 6 two-factor interactions, 4 three-factor interactions, and 1 four-factor interaction, whereas a $2^6$ experiment has 6 main effects, 15 two-factor interactions, 20 three-factor interactions, 15 four-factor interactions, 6 five-factor interactions, and 1 six-factor interaction. In most situations the **sparsity of effects principle** applies; that is, the system is usually dominated by the main effects and low-order interactions. Three-factor and higher interactions are usually negligible. Therefore, when the number of factors is moderately large—say, $k \geq 4$ or 5—a common practice is to run only a single replicate of the $2^k$ design and then pool or combine the higher-order interactions as an estimate of error.

## EXAMPLE 13.8 Characterizing a Plasma Etching Process

An article in *Solid State Technology* ("Orthogonal Design for Process Optimization and Its Application in Plasma Etching," May 1987, pp. 127–132) describes the application of factorial designs in developing a nitride etch process on a single-wafer plasma etcher. The process uses $C_2F_6$ as the reactant gas. It is possible to vary the gas flow, the power applied to the cathode, the pressure in the reactor chamber, and the spacing between the anode and the cathode (gap). Several response variables would usually be of interest in this process, but in this example we will concentrate on etch rate for silicon nitride. Perform an appropriate experiment to characterize the performance of this etching process with respect to the four process variables.

### SOLUTION

The authors used a single replicate of a $2^4$ design to investigate this process. Since it is unlikely that the three-factor and four-factor interactions are significant, we will tentatively plan to combine them as an estimate of error. The factor levels used in the design are shown here:

| Design Factor Level | Gap $A$ (cm) | Pressure $B$ (m Torr) | $C_2F_6$ Flow $C$ (SCCM) | Power $D$ (W) |
|---|---|---|---|---|
| Low (–) | 0.80 | 450 | 125 | 275 |
| High (+) | 1.20 | 550 | 200 | 325 |

Table 13.15 presents the data from the 16 runs of the $2^4$ design. The design is shown geometrically in Fig. 13.30. Table 13.16 is the table of plus and minus signs for the $2^4$ design. The signs in the columns of this table can be used to estimate the

(*continued*)

■ **TABLE 13.15**
**The $2^4$ Design for the Plasma Etch Experiment**

| Run | *A* (Gap) | *B* (Pressure) | *C* ($C_2F_6$ flow) | *D* (Power) | Etch Rate (Å/min) |
|---|---|---|---|---|---|
| 1 | −1 | −1 | −1 | −1 | 550 |
| 2 | 1 | −1 | −1 | −1 | 669 |
| 3 | −1 | 1 | −1 | −1 | 604 |
| 4 | 1 | 1 | −1 | −1 | 650 |
| 5 | −1 | −1 | 1 | −1 | 633 |
| 6 | 1 | −1 | 1 | −1 | 642 |
| 7 | −1 | 1 | 1 | −1 | 601 |
| 8 | 1 | 1 | 1 | −1 | 635 |
| 9 | −1 | −1 | −1 | 1 | 1037 |
| 10 | 1 | −1 | −1 | 1 | 749 |
| 11 | −1 | 1 | −1 | 1 | 1052 |
| 12 | 1 | 1 | −1 | 1 | 868 |
| 13 | −1 | −1 | 1 | 1 | 1075 |
| 14 | 1 | −1 | 1 | 1 | 860 |
| 15 | −1 | 1 | 1 | 1 | 1063 |
| 16 | 1 | 1 | 1 | 1 | 729 |

factor effects. To illustrate, the estimate of the effect of gap on factor *A* is

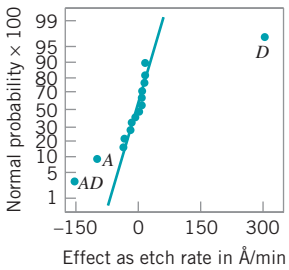
$$A = \frac{1}{8}[a + ab + ac + abc + ad + abd + acd + abcd - (1) - b - c - d - bc - bd - cd - bcd]$$
$$= \frac{1}{8}[669 + 650 + 642 + 635 + 749 + 868 + 860 + 729 - 550 - 604 - 633 - 601 - 1037 - 1052 - 1075 - 1063]$$
$$= -101.625$$

Thus, the effect of increasing the gap between the anode and the cathode from 0.80 cm to 1.20 cm is to decrease the etch rate by 101.625 angstroms per minute. It is easy to verify that the complete set of effect estimates is

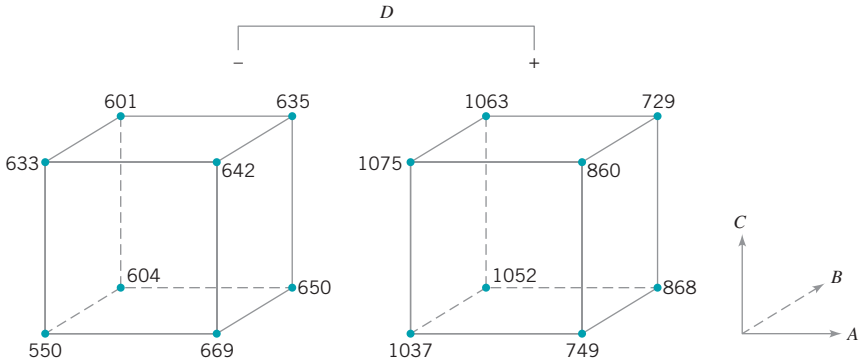
| | | | |
|---|---|---|---|
| $A =$ | −101.625 | $AD =$ | −153.625 |
| $B =$ | −1.625 | $BD =$ | −0.625 |
| $AB =$ | −7.875 | $ABD =$ | 4.125 |
| $C =$ | 7.375 | $CD =$ | −2.125 |
| $AC =$ | −24.875 | $ACD =$ | 5.625 |
| $BC =$ | −43.875 | $BCD =$ | −25.375 |
| $ABC =$ | −15.625 | $ABCD =$ | −40.125 |
| $D =$ | 306.125 | | |

A very helpful method in judging the significance of factors in a $2^k$ experiment is to construct a **normal probability plot** of the effect estimates. If none of the effects is significant, then the estimates will behave like a random sample drawn from a normal distribution with zero mean, and the plotted effects will lie approximately along a straight line. Those effects that do not plot on the line are significant factors.

The normal probability plot of effect estimates from the plasma etch experiment is shown in Fig. 13.31. Clearly, the main effects of *A* and *D* and the *AD* interaction are significant, as they fall far from the line passing through the other points. The analysis of variance summarized in Table 13.17 confirms these




■ **FIGURE 13.31**
Normal probability plot of effects, Example 13.8.




■ **FIGURE 13.30** The $2^4$ design for Example 13.8. The etch rate response is shown at the corners of the cubes.

**TABLE 13.16**
**Contrast Constants for the $2^4$ Design**

| Run | | *A* | *B* | *AB* | *C* | *AC* | *BC* | *ABC* | *D* | *AD* | *BD* | *ABD* | *CD* | *ACD* | *BCD* | *ABCD* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (1) | − | − | + | − | + | + | − | − | + | + | − | + | − | − | + |
| 2 | *a* | + | − | − | − | − | + | + | − | − | + | + | + | + | − | − |
| 3 | *b* | − | + | − | − | + | − | + | − | + | − | + | + | − | + | − |
| 4 | *ab* | + | + | + | − | − | − | − | − | − | − | − | + | + | + | + |
| 5 | *c* | − | − | + | + | − | − | + | − | + | + | − | − | + | + | − |
| 6 | *ac* | + | − | − | + | + | − | − | − | − | + | + | − | − | + | + |
| 7 | *bc* | − | + | − | + | − | + | − | − | + | − | + | − | + | − | + |
| 8 | *abc* | + | + | + | + | + | + | + | − | − | − | − | − | − | − | − |
| 9 | *d* | − | − | + | − | + | + | − | + | − | − | + | − | + | + | − |
| 10 | *ad* | + | − | − | − | − | + | + | + | + | − | − | − | − | + | + |
| 11 | *bd* | − | + | − | − | + | − | + | + | − | + | − | − | + | − | + |
| 12 | *abd* | + | + | + | − | − | − | − | + | + | + | + | − | − | − | − |
| 13 | *cd* | − | − | + | + | − | − | + | + | − | − | + | + | − | − | + |
| 14 | *acd* | + | − | − | + | + | − | − | + | + | − | − | + | + | − | − |
| 15 | *bcd* | − | + | − | + | − | + | − | + | − | + | − | + | − | + | − |
| 16 | *abcd* | + | + | + | + | + | + | + | + | + | + | + | + | + | + | + |

findings. Note that in the analysis of variance we have pooled the three- and four-factor interactions to form the error mean square. If the normal probability plot had indicated that any of these interactions were important, they would not be included in the error term.

Since $A = -101.625$, the effect of increasing the gap between the cathode and anode is to decrease the etch rate. However, $D = 306.125$, so applying higher power levels will increase the etch rate. Figure 13.32 is a plot of the *AD* interaction. This plot indicates that the effect of changing the gap width at low power settings is small, but that increasing the gap at high power settings dramatically reduces the etch rate. High etch rates are obtained at high power settings and narrow gap widths.

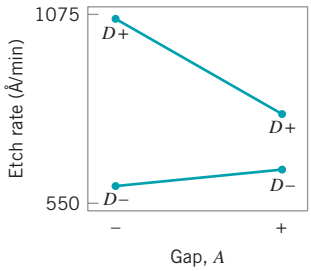



**FIGURE 13.32** *AD* interaction in the plasma etch experiment.

**TABLE 13.17**
**Analysis of Variance for the Plasma Etch Experiment**

| Source of Variation | Sum of Squares | Degrees of Freedom | Mean Square | $F_0$ |
|---|---|---|---|---|
| *A* | 41,310.563 | 1 | 41,310.563 | 20.28 |
| *B* | 10.563 | 1 | 10.563 | < 1 |
| *C* | 217.563 | 1 | 217.563 | < 1 |
| *D* | 374,850.063 | 1 | 374,850.063 | 183.99 |
| *AB* | 248.063 | 1 | 248.063 | < 1 |
| *AC* | 2,475.063 | 1 | 2,475.063 | 1.21 |
| *AD* | 94,402.563 | 1 | 99,402.563 | 46.34 |
| *BC* | 7,700.063 | 1 | 7,700.063 | 3.78 |
| *BD* | 1.563 | 1 | 1.563 | < 1 |
| *CD* | 18.063 | 1 | 18.063 | < 1 |
| Error | 10,186.815 | 5 | 2,037.363 | |
| Total | 531,420.938 | 15 | | |

(*continued*)

The regression model for this experiment is

$$\hat{y} = 776.0625 - \left(\frac{101.625}{2}\right)x_1 + \left(\frac{306.125}{2}\right)x_4 - \left(\frac{153.625}{2}\right)x_1x_4$$

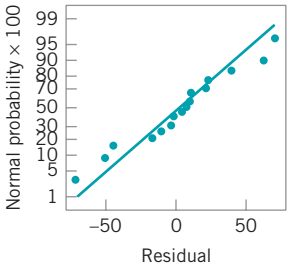
For example, when both $A$ and $D$ are at the low level, the predicted value from this model is

$$\hat{y} = 776.0625 - \left(\frac{101.625}{2}\right)(-1) + \left(\frac{306.125}{2}\right)(-1) - \left(\frac{153.625}{2}\right)(-1)(-1) = 597$$

The four residuals at this run are

$$e_1 = 550 - 597 = -47$$
$$e_2 = 604 - 597 = 7$$
$$e_3 = 633 - 597 = 36$$
$$e_4 = 601 - 597 = 4$$

The residuals at the other three runs, ($A$ high, $D$ low), ($A$ low, $D$ high), and ($A$ high, $D$ high), are obtained similarly. A normal probability plot of the residuals is shown in Fig. 13.33. The plot is satisfactory.




■ **FIGURE 13.33**
Normal probability plot of residuals, Example 13.8.

### 13.5.4 Addition of Center Points to the $2^k$ Design

A potential concern in the use of two-level factorial designs is the assumption of linearity in the factor effects. Of course, perfect linearity is unnecessary, and the $2^k$ system will work quite well even when the linearity assumption holds only approximately. In fact, we have already observed that when an interaction term is added to a main-effects model, curvature is introduced into the response surface. Since a $2^k$ design will support a main effects plus interactions model, some protection against curvature is already inherent in the design.

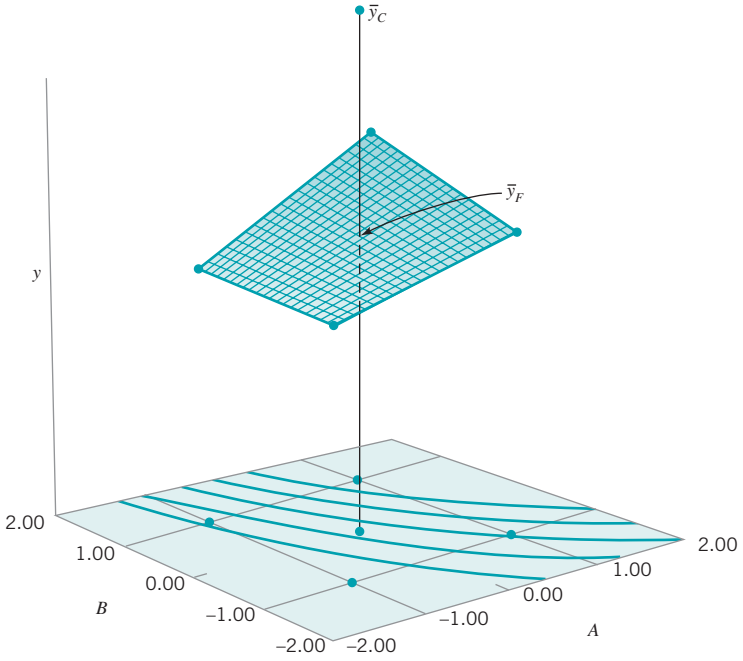
In some systems or processes, it will be necessary to incorporate **second-order effects** to obtain an adequate model. Consider the case of $k = 2$ factors. A model that includes second-order effects is

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{12}x_1x_2 + \beta_{11}x_1^2 + \beta_{22}x_2^2 + \varepsilon \qquad (13.27)$$

where the coefficients $\beta_{11}$ and $\beta_{22}$ measure pure quadratic effects. Equation (13.27) is a **second-order response surface model.** This model cannot be fitted using a $2^2$ design, because to fit a quadratic model, all factors must be run at at least three levels. It is important, however, to be able to determine whether the pure quadratic terms in equation (13.27) are needed.

There is a method of adding one point to a $2^k$ factorial design that will provide some protection against pure quadratic effects (in the sense that one can test to determine if the quadratic terms are necessary). Furthermore, if this point is replicated, then an independent estimate of experimental error can be obtained. The method consists of adding **center points** to the $2^k$ design. These center points consist of $n_C$ replicates run at the point $x_i = 0$ ($i = 1, 2, \ldots, k$). One important reason for adding the replicate runs at the design center is that center points do not impact the usual effect estimates in a $2^k$ design. We assume that the $k$ factors are quantitative; otherwise, a "middle" or center level of the factor would not exist.

To illustrate the approach, consider a $2^2$ design with one observation at each of the factorial points (−, −), (+, −), (−, +), and (+, +) and $n_C$ observations at the center points




■ **FIGURE 13.34** A $2^2$ design with center points.

(0, 0). Figure 13.34 illustrates the situation. Let $\bar{y}_F$ be the average of the four runs at the four factorial points, and let $\bar{y}_C$ be the average of the $n_C$ runs at the center point. If the difference $\bar{y}_F - \bar{y}_C$ is small, then the center points lie on or near the plane passing through the factorial points, and there is no curvature. On the other hand, if $\bar{y}_F - \bar{y}_C$ is large, then curvature is present. *A* single-degree-of-freedom sum of squares for pure quadratic curvature is given by

$$SS_{\text{pure quadratic}} = \frac{n_F n_C \left(\bar{y}_F - \bar{y}_C\right)^2}{n_F + n_C} \tag{13.28}$$

where, in general, $n_F$ is the number of factorial design points. This quantity may be compared to the error mean square to test for curvature.[1] More specifically, when points are added to the center of the $2^k$ design, then the model we may entertain is

$$y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \sum\sum_{i<j} \beta_{ij} x_i x_j + \sum_{j=1}^{k} \beta_{jj} x_j^2 + \varepsilon$$

where the $\beta_{jj}$ are pure quadratic effects. The test for curvature actually tests the hypotheses

$$H_0: \sum_{j=1}^{k} \beta_{jj} = 0$$

$$H_1: \sum_{j=1}^{k} \beta_{jj} \neq 0$$

Furthermore, if the factorial points in the design are unreplicated, we may use the $n_C$ center points to construct an estimate of error with $n_C - 1$ degrees of freedom.

[1] A *t*-test can also be used; see Section S13.2 of the supplemental text material for this chapter.

## EXAMPLE 13.9 Adding Center Points to a $2^k$ Design

Table 13.18 presents a modified version of the original unreplicated $2^4$ design in Example 13.8 to which $n_C = 4$ center points have been added. Analyze the data and draw conclusions.

## SOLUTION

The average of the center points is $\bar{y}_C = 752.75$ and the average of the 16 factorial points is $\bar{y}_F = 776.0625$. The curvature sum of squares is computed from equation (13.28) as

$$SS_{\text{pure quadratic}} = \frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{n_F + n_C} = \frac{16(4)(776.0625 - 752.75)^2}{16 + 4} = 1739.1$$

Furthermore, an estimate of experimental error can be obtained by simply calculating the sample variance of the four center points as follows:

$$\hat{\sigma}^2 = \frac{\sum_{i=17}^{20} (y_i - 752.75)^2}{3} = \frac{3122.7}{3} = 1040.92$$

This estimate of error has $n_C - 1 = 4 - 1 = 3$ degrees of freedom.

The pure quadratic sum of squares and the estimate of error may be incorporated into the analysis of variance for this experimental design. We would still use a normal probability plot of the effect estimates to preliminarily identify the important factors. The construction of this plot would not be affected by the addition of center points in the design, we would still identify *A* (Gap), *D* (Power), and the *AD* interaction as the most important effect.

Table 13.19 is the analysis of variance for this experiment obtained from Minitab. In the analysis, we included all four main effects and all six two-factor interactions in the model (just as we did in Example 13.8; see also Table 13.17). Note also that the pure quadratic sum of squares from equation

■ **TABLE 13.18**
**The $2^4$ Design for the Plasma Etch Experiment**

| Run | *A* (Gap) | *B* (Pressure) | *C* ($C_2F_6$ flow) | *D* (Power) | Etch Rate (Å/min) |
|---|---|---|---|---|---|
| 1 | −1 | −1 | −1 | −1 | 550 |
| 2 | 1 | −1 | −1 | −1 | 669 |
| 3 | −1 | 1 | −1 | −1 | 604 |
| 4 | 1 | 1 | −1 | −1 | 650 |
| 5 | −1 | −1 | 1 | −1 | 633 |
| 6 | 1 | −1 | 1 | −1 | 642 |
| 7 | −1 | 1 | 1 | −1 | 601 |
| 8 | 1 | 1 | 1 | −1 | 635 |
| 9 | −1 | −1 | −1 | 1 | 1037 |
| 10 | 1 | −1 | −1 | 1 | 749 |
| 11 | −1 | 1 | −1 | 1 | 1052 |
| 12 | 1 | 1 | −1 | 1 | 868 |
| 13 | −1 | −1 | 1 | 1 | 1075 |
| 14 | 1 | −1 | 1 | 1 | 860 |
| 15 | −1 | 1 | 1 | 1 | 1063 |
| 16 | 1 | 1 | 1 | 1 | 729 |
| 17 | 0 | 0 | 0 | 0 | 706 |
| 18 | 0 | 0 | 0 | 0 | 764 |
| 19 | 0 | 0 | 0 | 0 | 780 |
| 20 | 0 | 0 | 0 | 0 | 761 |

**TABLE 13.19**
**Analysis of Variance Output from Minitab for Example 13.9**

Estimated Effects and Coefficients for Etch Rate (coded units)

| Term | Effect | Coef | SE Coef | T | P |
|---|---|---|---|---|---|
| Constant | | 776.06 | 10.20 | 76.11 | 0.000 |
| A | -101.62 | -50.81 | 10.20 | -4.98 | 0.001 |
| B | -1.63 | -0.81 | 10.20 | -0.08 | 0.938 |
| C | 7.37 | 3.69 | 10.20 | 0.36 | 0.727 |
| D | 306.12 | 153.06 | 10.20 | 15.01 | 0.000 |
| A*B | -7.88 | -3.94 | 10.20 | -0.39 | 0.709 |
| A*C | -24.88 | -12.44 | 10.20 | -1.22 | 0.257 |
| A*D | -153.63 | -76.81 | 10.20 | -7.53 | 0.000 |
| B*C | -43.87 | -21.94 | 10.20 | -2.15 | 0.064 |
| B*D | -0.63 | -0.31 | 10.20 | -0.03 | 0.976 |
| C*D | -2.13 | -1.06 | 10.20 | -0.10 | 0.920 |
| Ct Pt | | -23.31 | 22.80 | -1.02 | 0.337 |

Analysis of Variance for Etch (coded units)

| Source | DF | Seq SS | Adj SS | Adj MS | F | P |
|---|---|---|---|---|---|---|
| Main Effects | 4 | 416389 | 416389 | 104097 | 62.57 | 0.000 |
| 2-Way Interactions | 6 | 104845 | 104845 | 17474 | 10.50 | 0.002 |
| Curvature | 1 | 1739 | 1739 | 1739 | 1.05 | 0.337 |
| Residual Error | 8 | 13310 | 13310 | 1664 | | |
| Lack of Fit | 5 | 10187 | 10187 | 2037 | 1.96 | 0.308 |
| Pure Error | 3 | 3123 | 3123 | 1041 | | |
| Total | 19 | 536283 | | | | |

13.28 is called the "curvature" sum of squares, and the estimate of error calculated from the $n_C = 4$ center points is called the "pure error" sum of squares in Table 13.19. The "lack-of-fit" sum of squares in Table 13.19 is actually the total of the sums of squares for the three-factor and four-factor interactions. The $F$-test for lack of fit is computed as

$$F_0 = \frac{MS_{\text{lack of fit}}}{MS_{\text{pure error}}} = \frac{2037}{1041} = 1.96$$

and is not significant, indicating that none of the higher-order interaction terms is important. This computer program combines the pure error and lack-of-fit sum of squares to form a residual sum of squares with eight degrees of freedom. This residual sum of squares is used to test for pure quadratic curvature with

$$F_0 = \frac{MS_{\text{curvature}}}{MS_{\text{residual}}} = \frac{1739}{1664} = 1.05$$

The $P$-value in Table 13.19 associated with this $F$-ratio indicates that there is no evidence of pure quadratic curvature.

The upper portion of Table 13.19 shows the regression coefficient for each model effect, the corresponding $t$-value, and the $P$-value. Clearly the main effects of $A$ and $D$ and the $AD$ interaction are the three largest effects.

### 13.5.5 Blocking and Confounding in the $2^k$ Design

It is often impossible to run all of the observations in a $2^k$ factorial design under constant or homogeneous conditions. For example, it might not be possible to conduct all the tests on one shift or use material from a single batch. When this problem occurs, **blocking** is an excellent technique for eliminating the unwanted variation that could be caused by the nonhomogeneous conditions. If the design is replicated, and if the block is of sufficient size, then one approach

is to run each replicate in one **block** (set of homogeneous conditions). For example, consider a $2^3$ design that is to be replicated twice. Suppose that it takes about 1 h to complete each run. Then by running the eight runs from replicate one on one day and the eight runs from the second replicate on another day, any time effect, or difference between how the process works on the two days, can be eliminated. Thus, the two days became the two blocks in the design. The average difference between the responses on the two days is the block effect. For more introductory material about blocking and how it is used to eliminate the effect of a nuisance factor, see Section S13.3 of the supplemental material for this chapter.

Sometimes we cannot run a complete replicate of a factorial design under homogeneous experimental conditions. **Confounding** is a design technique for running a factorial experiment in blocks, where the block size is smaller than the number of runs in one complete replicate. The technique causes certain interactions to be indistinguishable from or **confounded with blocks.** We will illustrate confounding in the $2^k$ factorial design in $2^p$ blocks, where $p < k$.

Consider a $2^2$ design. Suppose that each of the $2^2 = 4$ runs require 4 h of laboratory analysis. Thus, two days are required to perform the experiment. If days are considered as blocks, then we must assign two of the four runs to each day.

Consider the design shown in Fig. 13.35. Note that block 1 contains the runs (1) and *ab* and that block 2 contains *a* and *b*. The contrasts for estimating the main effects *A* and *B* are

$$\text{Contrast}_A = ab + a - b - (1)$$
$$\text{Contrast}_B = ab + b - a - (1)$$

These contrasts are unaffected by blocking since in each contrast there is one plus and one minus run from each block. That is, any difference between block 1 and block 2 will cancel out. The contrast for the *AB* interaction is

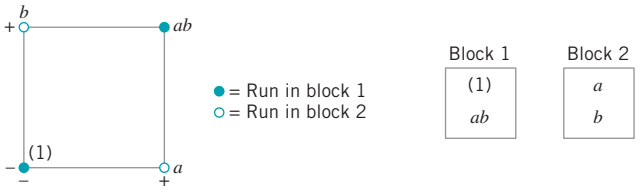
$$\text{Contrast}_{AB} = ab + (1) - a - b$$

Since the two runs with the plus sign, *ab* and (1), are in block 1 and the two with the minus sign, *a* and *b*, are in block 2, the block effect and *AB* interaction are identical. That is, *AB* is confounded with blocks.

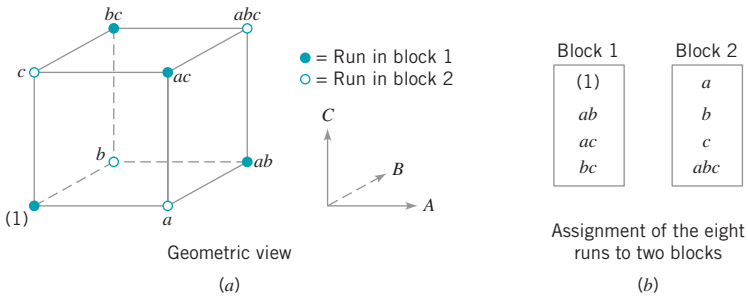
The reason for this is apparent from the table of plus and minus signs for the $2^2$ design, shown in Table 13.7. From this table, we see that all runs that have a plus on *AB* are assigned to block 1, and all runs that have a minus sign on *AB* are assigned to block 2.

This scheme can be used to confound any $2^k$ design in two blocks. As a second example, consider a $2^3$ design, run in two blocks. Suppose we wish to confound the three-factor interaction *ABC* with blocks. From the table of plus and minus signs, shown in Table 13.11, we assign the runs that are minus on *ABC* to block 1 and those that are plus on *ABC* to block 2. The resulting design is shown in Fig. 13.36.

For more information on confounding, refer to Montgomery (2005, Chapter 8). This book contains guidelines for selecting factors to confound with blocks so that main effects




■ **FIGURE 13.35** A $2^2$ design in two blocks.




■ **FIGURE 13.36** The $2^3$ design in two blocks with $ABC$ confounded.

and low-order interactions are not confounded. In particular, the book contains a table of suggested confounding schemes for designs with up to seven factors and a range of block sizes, some as small as two runs.

## 13.6 Fractional Replication of the $2^k$ Design

As the number of factors in a $2^k$ design increases, the number of runs required increases rapidly. For example, a $2^5$ requires 32 runs. In this design, only 5 degrees of freedom correspond to main effects and 10 degrees of freedom correspond to two-factor interactions. If we can assume that certain high-order interactions are negligible, then a fractional factorial design involving fewer than the complete set of $2^k$ runs can be used to obtain information on the main effects and low-order interactions. In this section, we will introduce fractional replication of the $2^k$ design. For a more complete treatment, see Montgomery (2005, Chapter 8).

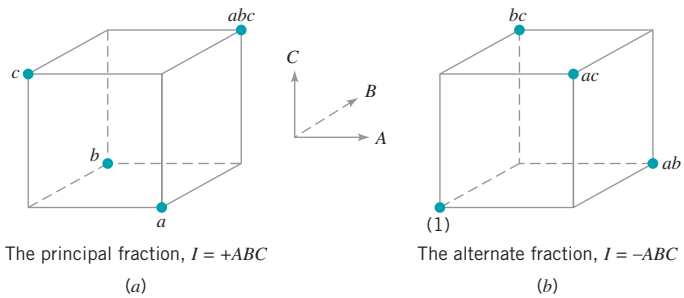
### 13.6.1 The One-Half Fraction of the $2^k$ Design

A one-half fraction of the $2^k$ design contains $2^{k-1}$ runs and is often called a $2^{k-1}$ fractional factorial design. As an example, consider the $2^{3-1}$ design—that is, a one-half fraction of the $2^3$. The table of plus and minus signs for the $2^3$ design is shown in Table 13.20. Suppose we select the four runs $a$, $b$, $c$, and $abc$ as our one-half fraction. These runs are shown in the top half of Table 13.20. The design is shown geometrically in Fig. 13.37*a*.

Note that the $2^{3-1}$ design is formed by selecting only those runs that yield a plus on the $ABC$ effect. Thus, $ABC$ is called the **generator** of this particular fraction. Furthermore, the identity element $I$ is also plus for the four runs, so we call

$$I = ABC$$

the **defining relation** for the design.




■ **FIGURE 13.37** The one-half fractions of the $2^3$ design.